

HG-MS

Magnetic Contactors B Type

Magnetic Contactors & Thermal Overload Relays

HD HYUNDAI ELECTRIC

Essential for Today, Potential for Tomorrow

HD Hyundai Electric solely pursues the growth of our customers' business. From power generation to power distribution, we focus on developing and commercializing products and solutions aimed at increasing the efficiency of energy equipment as well as at proactively monitoring and controlling assets in an integrated manner to improve our customers' productivity and management efficiency. We are well aware that our efforts add to the driving force behind our customers' growth and contribute to the creation and maintenance of a more dynamic world. We focus on achieving innovation and strive to evolve continuously to shape a better tomorrow based on today's technological advancement

Solution

INTEGRICT

Energy Solution

Energy solution business refers to the business of designing, procuring and establishing a system that enables the efficient use of power energy through integrated management of the production, consumption, sales and operation.

Asset Management Solution

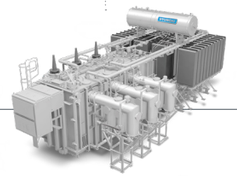
Asset management solution is a business that maximizes the overall business efficiency by systematically managing the performance, risk, maintenance cost and others as well as by providing an asset management solution suitable to the customer's circumstance depending on the product lifecycle (PLC) of various products.

Utility

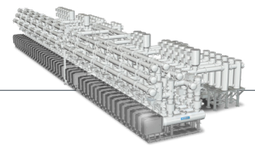
Generation(step-up)

Power Plants

- Supplied to more than 70 countries around the world for the past 45 over years since 1978
- Satisfies the various demands of customers through the acquisition of quality certifications from international accredited institute
- Participates in the world's key technical committee such as GIGRE and others, pioneering the establishment of technology standard related to power network



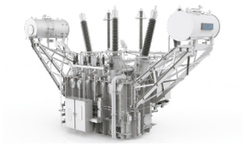
Power Transformer
· up to 800 kV, 1,500 MVA



Gas Insulated Switchgear
· up to 800 kV

- Can be installed in spaces smaller than the open type of substation by using SF_6 gas with outstanding insulation and arc extinguishing characteristics
- Manufacturing advanced products that are resistant to external environment and climate effects through the sealing at the charge part
- Extensive project experiences around the world
- Reduces installation period and cost due to simple installation and transportation, convenient maintenance
- Design considering the safety of the workers as priority

GREENTRIC



ECO Transformer
· up to 500 kV, 1,000 MVA

- Eco-friendly oil (Natural/Synthetic)
- Dynamic Pressure Resistant System(DPRS)
- Guarantees excellence in safety

GREENTRIC



SF_6 Free Gas Insulated Switchgear
· GIS for 170 kV 50 kA 60 Hz
· ECO GIS for 72.5 kV, 145 kV

- Korea's first solely developed technology
- Eco-friendly Novec mixture gas instead of SF_6 gas

Transmission

Primary Substation/Secondary Substation



Gas Insulated Switchgear
· up to 800 kV



Power Transformer
· up to 800 kV, 1,500 MVA



Shunt Reactor
· up to 765 kV, 300 Mvar

Distribution



Cubicle GIS
· up to 40.5 kV
· IEC, ANSI, GOST, UL

- Produces high quality products using angle-less type
- Multi-functional digital protection relay (HiMAP) applied
- High reliability secured, provides various operation information such as protection, measurement and control
- Firm external box, size and compact, making it safe
- Maintains high quality through stringent quality control system and continuous research and development

Motor & Generator



Synchronous Generator
· Rated Output: < 50 MVA
· Rated Voltage: < 15 kV
· Ploes: > 2P

Marine

Marine Solution

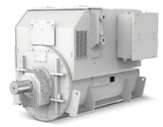
- Production of high quality marine devices satisfying the regulations and standards of key marine associations (LRS, ABS, DNV, GL, BV, NK etc.) and world's renowned institutes
- High quality safety secured through the latest equipment and stringent quality control system
- Realization of optimal high efficiency by converging SWGR, Generator, Motor, Telecom, Automation, Intelligent Motor Control Unit and others



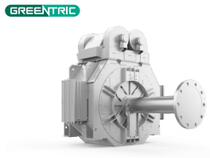
Marine Switchgear



Marine Motor



Generator



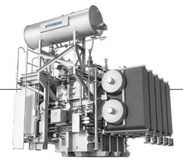
Shaft Generator



Metal Clad Switchgear
· up to 38 kV
· IEC, ANSI, UL



Low Voltage Switchgear & Motor Control Center
· H8PU : 480 V, 3,200 A, 65 kA
· H5600 : 480 V, 1,200 A, 65 kA
· HiMCC : 480 V, 5,000 A, 100 kA



Distribution Transformer
· up to 36 kV, 50 MVA



Cast Resin Transformer
· up to 36 kV, 20 MVA



Synchronous Motor
· Rated Output: < 40 MW
· Rated Voltage: < 15 kV
· Ploes: > 2P



MV&HV Induction Motor
· Rated Output: < 35 MW
· Rated Voltage: < 15 kV
· Poles: 2P-30P



VCB
· IEC, ANSI, UL
· up to 36/38 kV, 50 kA, 4,000 A



ACB
· AC : up to 150 kA, 6,300 A
up to 130 kA, 6,000 A
(UL1066, ANSI C37.13/16/17/50, CSA C22.2 no. 31)
· DC : up to 60 kA, 4,000 A



VC
· up to 12 kV, 400 A



Switch-Disconnector
· DC : up to 100 kA/1 s, 4,000 A



HGMAP
· HGMAP Series



HGCAM
· HGCAM Series



MCCB
· AC : up to 150 kA, 1,600 A
· DC : up to 100 kA, 800 A



ELCB
· up to 85 kA, 800 A, 1,000 mA



MCCB / ELCB
· up to 10 kA, 100 A, 100 mA



MS / MC
· up to 800 A



Distribution Box
· Up to 36 ways
· Surface/Flush type



MCB
· up to 15 kA, 125 A



RCCB / RCBO
· up to 100 A, 500 mA



Isolator
· up to 125 A



Electronic Circuit Breaker
· up to 20 A



MMS
· up to 100 kA, 80 A



Fuse Links & Switches
· up to 1,250 A

- Can be applied to various places of demand due to various domestic and international certifications and marine certifications
- Full HG-Series line-up established
- High standard of breaking capacity and various product line offers production in any load environments
- Increased reliability with reinforced breaking performance of high-breaking MCCB
- Reinforced usage stability through internal reliability verification



Installation Contactor
· up to 63 A



SPD
· up to 200 kA
· AC, DC



HGDM



EOCR
· up to 60 A

- Enhanced reliability and secured safety with production of products based on the world's best equipment and stringent quality system
- Realized high efficiency by selecting slot based on FEM
- Realized small and lightweight with optimal design based on FEM analysis method
- Satisfies the quality standards of international accredited institutes (IEC, IEEE, CSA, NEMA, API etc.)



LV Motor
· Rated Output: < 750 kW
· Rated Voltage: < 690 V
· Poles: 2P-12P

HG
Magnetic Contactors & Thermal Overload Relays

HGC B Type	Overview and Characteristics	2
	Model Selection Table	4
	Accessories	14
	Technical Data	20
	Order Code	21
HGT B Type	Overview and Characteristics	22
	Model Selection Table	24
	Technical Data	27
	Order Code	29
Dimensions		30
Handling and Maintenance Inspection		35
Acquired Standards		38

MS

HG Series

Magnetic Contactors

Motor protection control solutions for various industrial facilities!



Can be installed with Din rail and Screw

Various accessories

Increase customer convenience & safety

Overload Relays(HGT B Type) can be used directly with Magnetic contactor (HGC B Type) products

Designed to prevent physical contact with energized parts

IEC International Standards

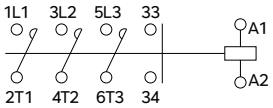
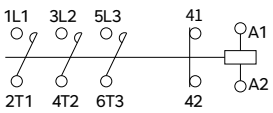
IEC60947-4-1 application



Magnetic contactor (HGC B Type)

Model Selection Table

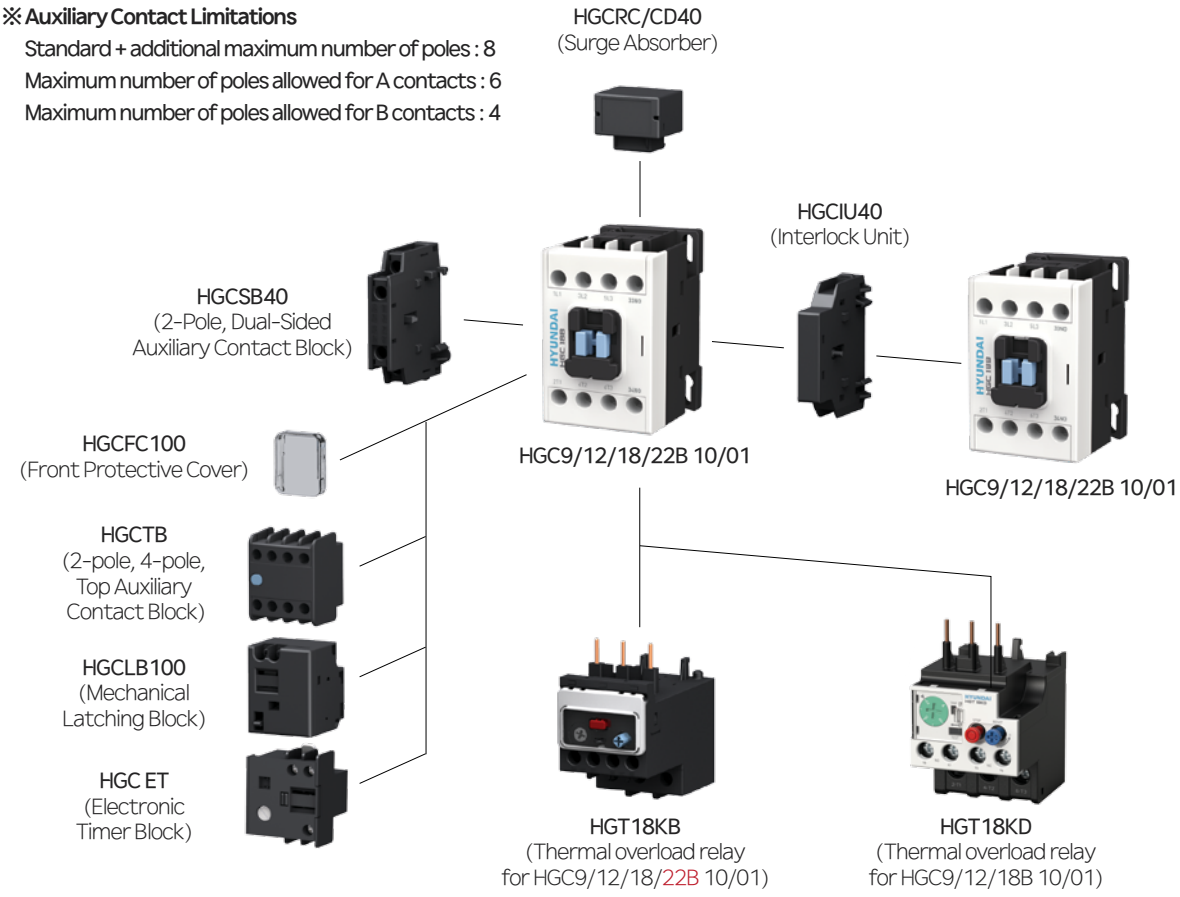
Magnetic Contractor : HGC9/12/18/22B 10 (1a), HGC9/12/18/22B 01 (1b)

Rating				HGC			
Model			**1a Type 1b Type**	**HGC9B 10 HGC9B 01**	**HGC12B 10 HGC12B 01**	**HGC18B 10 HGC18B 01**	**HGC22B 10 HGC22B 01**
IEC60947-4-1							
Pole			P	3	3	3	3
Rated Frequency			Hz	50/60	50/60	50/60	50/60
Rated Insulation Voltage [Ui]			V	800	800	800	800
Rated Operational Voltage [Ue]			V	690	690	690	690
Rated Impulse Withstand Voltage [Uimp]			kV	6	6	6	6
Rated Thermal Current [Ith] (AC1)			A	25	30	40	45
AC3	200 ~ 240V		kW/A	2.5/9	3.5/12	4.5/18	5.5/22
	380 ~ 440V			4/9	5.5/12	7.5/18	11/22
	500 ~ 550V			4/7	7.5/12	8.5/13	15/22
	660 ~ 690V			4/6	7.5/9	7.5/9	15/18
	Lifespan	Electrical	10,000 Times	140	140	140	140
		Mechanical		1,000	1,000	1,000	1,000
AC4	200 ~ 240V		kW/A	1.5/8	2.2/11	3.7/16	3.7/18
	380 ~ 440V			2.2/6	4/9	4/11	5.5/13
	Electrical Lifespan		10,000 Times	3	3	3	3
AC1, 2, 3 Load Permitted Operational Frequency/Time	100% Load		Frequency	1,000	1,000	1,000	1,000
	50% Load (DC)			2,000	2,000	2,000	2,000
	20% Load (DC)			3,600	3,600	3,600	3,600
AC4 Load Permitted Operational Frequency/Time	100% Load		Frequency	300	300	250	250
	50% Load			600	600	500	500
Making Capacity	220V		A	110	130	180	220
	440V			90	120	180	220
Breaking Capacity	220V		A	88	104	144	176
	440V			72	96	144	176
Mounting Method				Screw & Din Rail Mounting			
Standard Auxiliary Contacts				1a or 1b			
External Dimensions	W×H×D		mm	45×75×86			
Weight			kg	0.35			
Contact Arrangement							
HGC9/12/18/22B 10 (1a type)	Main Contact		3a				
	Auxiliary contact		1a				
HGC9/12/18/22B 01 (1b type)	Main Contact		3a				
	Auxiliary contact		1b				

Accessories

※ Auxiliary Contact Limitations

Standard + additional maximum number of poles : 8
Maximum number of poles allowed for A contacts : 6
Maximum number of poles allowed for B contacts : 4

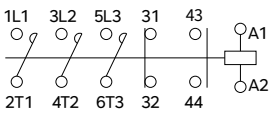


Order Code

Operational Voltage (V)		1a Type				1b Type			
		HGC9B 10	HGC12B 10	HGC18B 10	HGC22B 10	HGC9B 01	HGC12B 01	HGC18B 01	HGC22B 01
AC (50Hz)	24	HGC9B 10NS X024	HGC12B 10NS X024	HGC18B 10NS X024	HGC22B 10NS X024	HGC9B 01NS X024	HGC12B 01NS X024	HGC18B 01NS X024	HGC22B 01NS X024
	110	HGC9B 10NS X110	HGC12B 10NS X110	HGC18B 10NS X110	HGC22B 10NS X110	HGC9B 01NS X110	HGC12B 01NS X110	HGC18B 01NS X110	HGC22B 01NS X110
	220	HGC9B 10NS X220	HGC12B 10NS X220	HGC18B 10NS X220	HGC22B 10NS X220	HGC9B 01NS X220	HGC12B 01NS X220	HGC18B 01NS X220	HGC22B 01NS X220
	240	HGC9B 10NS X240	HGC12B 10NS X240	HGC18B 10NS X240	HGC22B 10NS X240	HGC9B 01NS X240	HGC12B 01NS X240	HGC18B 01NS X240	HGC22B 01NS X240
	380	HGC9B 10NS X380	HGC12B 10NS X380	HGC18B 10NS X380	HGC22B 10NS X380	HGC9B 01NS X380	HGC12B 01NS X380	HGC18B 01NS X380	HGC22B 01NS X380
	440	HGC9B 10NS X440	HGC12B 10NS X440	HGC18B 10NS X440	HGC22B 10NS X440	HGC9B 01NS X440	HGC12B 01NS X440	HGC18B 01NS X440	HGC22B 01NS X440
AC (60Hz)	24	HGC9B 10NS A024	HGC12B 10NS A024	HGC18B 10NS A024	HGC22B 10NS A024	HGC9B 01NS A024	HGC12B 01NS A024	HGC18B 01NS A024	HGC22B 01NS A024
	110	HGC9B 10NS A110	HGC12B 10NS A110	HGC18B 10NS A110	HGC22B 10NS A110	HGC9B 01NS A110	HGC12B 01NS A110	HGC18B 01NS A110	HGC22B 01NS A110
	220	HGC9B 10NS A220	HGC12B 10NS A220	HGC18B 10NS A220	HGC22B 10NS A220	HGC9B 01NS A220	HGC12B 01NS A220	HGC18B 01NS A220	HGC22B 01NS A220
	240	HGC9B 10NS A240	HGC12B 10NS A240	HGC18B 10NS A240	HGC22B 10NS A240	HGC9B 01NS A240	HGC12B 01NS A240	HGC18B 01NS A240	HGC22B 01NS A240
	380	HGC9B 10NS A380	HGC12B 10NS A380	HGC18B 10NS A380	HGC22B 10NS A380	HGC9B 01NS A380	HGC12B 01NS A380	HGC18B 01NS A380	HGC22B 01NS A380
	440	HGC9B 10NS A440	HGC12B 10NS A440	HGC18B 10NS A440	HGC22B 10NS A440	HGC9B 01NS A440	HGC12B 01NS A440	HGC18B 01NS A440	HGC22B 01NS A440

Model Selection Table

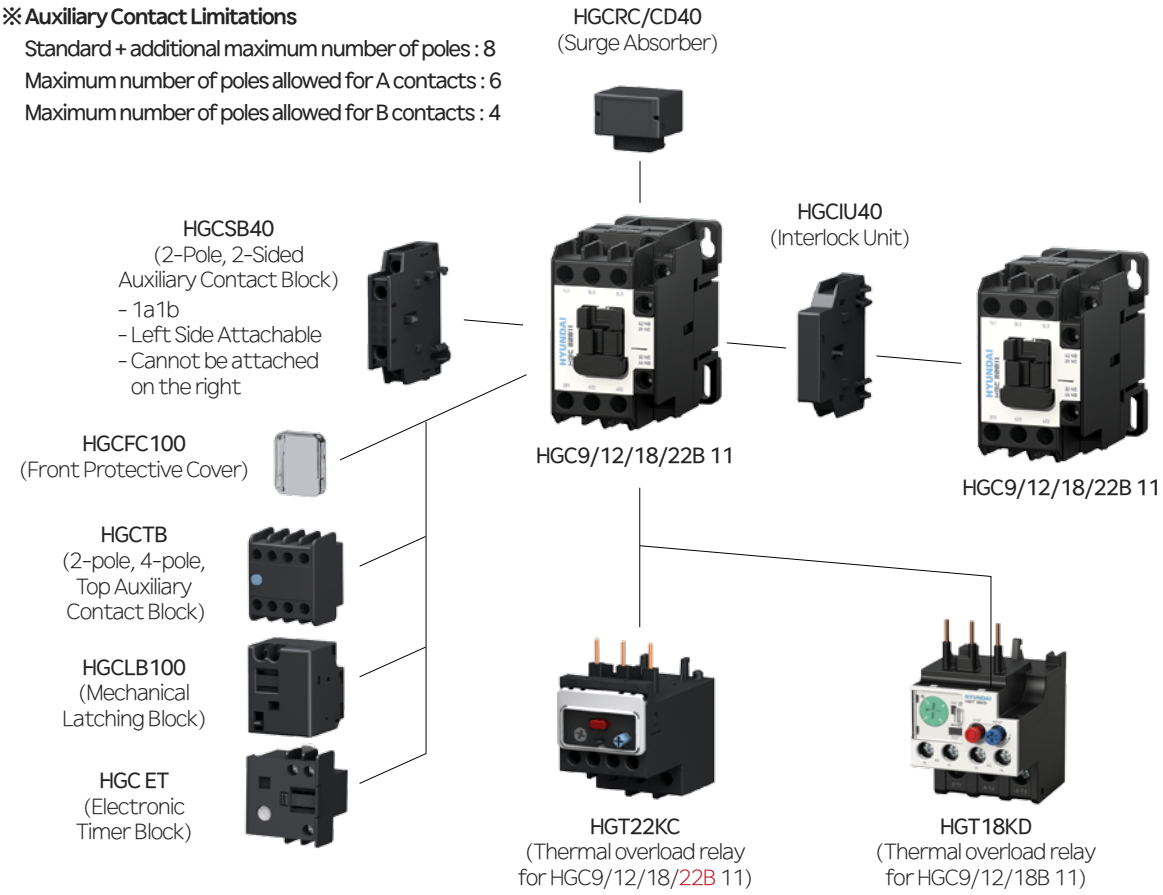
Magnetic Contactor : HGC9/12/18/22B 11 (1a1b)

Rating				HGC			
Model				**HGC9B11**	**HGC12B11**	**HGC18B11**	**HGC22B11**
IEC60947-4-1							
Pole			P	3	3	3	3
Rated Frequency			Hz	50/60	50/60	50/60	50/60
Rated Insulation Voltage [Ui]			V	690	690	690	690
Rated Operational Voltage [Ue]			V	690	690	690	690
Rated Impulse Withstand Voltage [Uimp]			kV	6	6	6	6
Rated Thermal Current [Ith] (AC1)			A	25	30	40	45
AC3	200 ~ 240V		kW/A	2.5/9	3.5/12	4.5/18	5.5/22
	380 ~ 440V			4/9	5.5/12	7.5/18	11/22
	500 ~ 550V			4/7	7.5/12	8.5/13	15/22
	660 ~ 690V			4/6	7.5/9	7.5/9	15/18
	Lifespan	Electrical	10,000 Times	140	140	140	140
		Mechanical		1,000	1,000	1,000	1,000
AC4	200 ~ 240V		kW/A	1.5/8	2.2/11	3.7/16	3.7/18
	380 ~ 440V			2.2/6	4/9	4/11	5.5/13
	Electrical Lifespan		10,000 Times	3	3	3	3
AC1, 2, 3 Load Permitted Operational Frequency/Time	100% Load		Frequency	1,000	1,000	1,000	1,000
	50% Load (DC)			2,000	2,000	2,000	2,000
	20% Load (DC)			3,600	3,600	3,600	3,600
AC4 Load Permitted Operational Frequency/Time	100% Load		Frequency	300	300	250	250
	50% Load			600	600	500	500
Making Capacity	220V		A	110	130	180	220
	440V			90	120	180	220
Breaking Capacity	220V		A	88	104	144	176
	440V			72	96	144	176
Mounting Method				Screw & Din Rail Mounting			
Standard Auxiliary Contacts				1a1b			
External Dimensions	W×H×D		mm	45×75.8×84.4			
Weight			kg	0.4			
Contact Arrangement							
Standard	Main Contact		3a				
	Auxiliary Contact		1a1b				

Accessories

※ Auxiliary Contact Limitations

Standard + additional maximum number of poles : 8
Maximum number of poles allowed for A contacts : 6
Maximum number of poles allowed for B contacts : 4

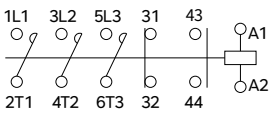


Order Code

Operational Voltage (V)		HGC9B	HGC12B	HGC18B	HGC22B
AC (50Hz)	24	HGC9B 11NS X024	HGC12B 11NS X024	HGC18B 11NS X024	HGC22B 11NS X024
	110	HGC9B 11NS X110	HGC12B 11NS X110	HGC18B 11NS X110	HGC22B 11NS X110
	220	HGC9B 11NS X220	HGC12B 11NS X220	HGC18B 11NS X220	HGC22B 11NS X220
	240	HGC9B 11NS X240	HGC12B 11NS X240	HGC18B 11NS X240	HGC22B 11NS X240
	380	HGC9B 11NS X380	HGC12B 11NS X380	HGC18B 11NS X380	HGC22B 11NS X380
	440	HGC9B 11NS X440	HGC12B 11NS X440	HGC18B 11NS X440	HGC22B 11NS X440
AC (60Hz)	24	HGC9B 11NS A024	HGC12B 11NS A024	HGC18B 11NS A024	HGC22B 11NS A024
	110	HGC9B 11NS A110	HGC12B 11NS A110	HGC18B 11NS A110	HGC22B 11NS A110
	220	HGC9B 11NS A220	HGC12B 11NS A220	HGC18B 11NS A220	HGC22B 11NS A220
	240	HGC9B 11NS A240	HGC12B 11NS A240	HGC18B 11NS A240	HGC22B 11NS A240
	380	HGC9B 11NS A380	HGC12B 11NS A380	HGC18B 11NS A380	HGC22B 11NS A380
	440	HGC9B 11NS A440	HGC12B 11NS A440	HGC18B 11NS A440	HGC22B 11NS A440

Model Selection Table

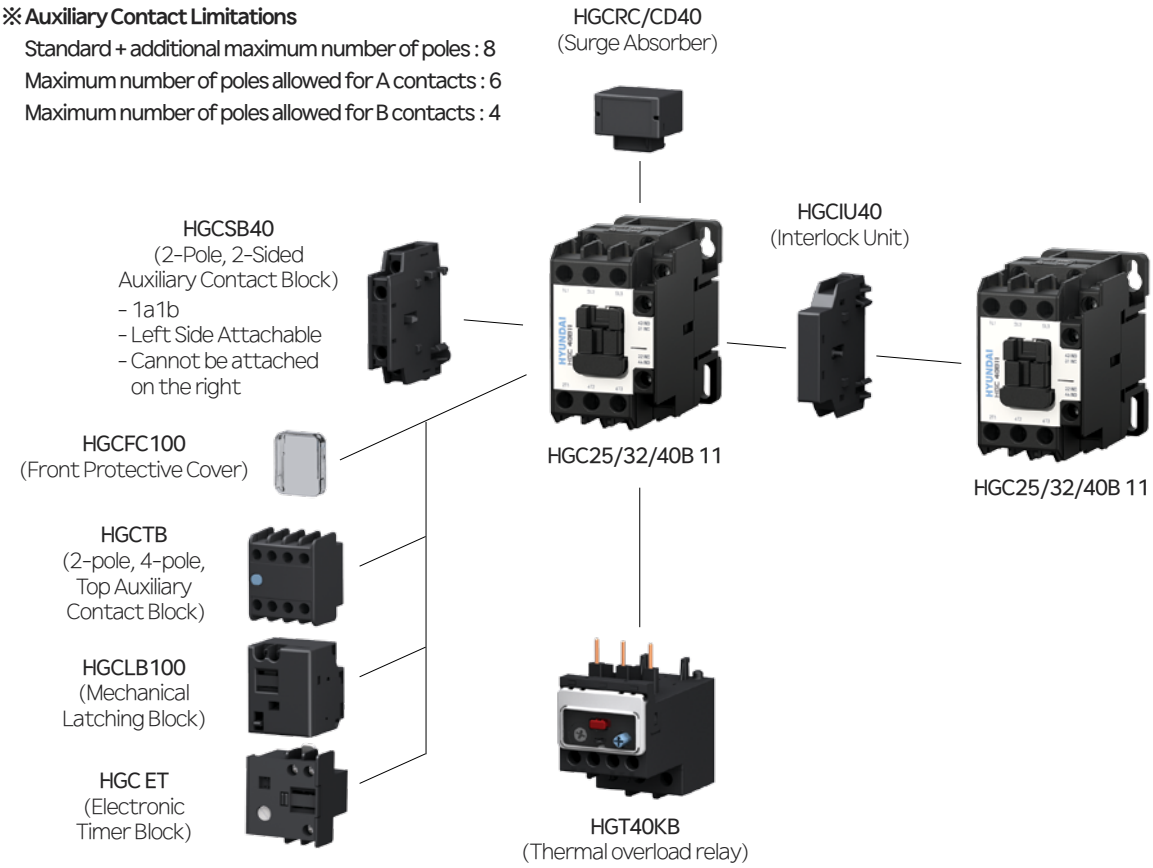
Magnetic Contactor : HGC25/32/40B 11 (1a1b)

Rating				HGC		
Model				**HGC25B11**	**HGC32B11**	**HGC40B11**
IEC60947-4-1						
Pole			P	3	3	3
Rated Frequency			Hz	50/60	50/60	50/60
Rated Insulation Voltage [Ui]			V	690	690	690
Rated Operational Voltage [Ue]			V	690	690	690
Rated Impulse Withstand Voltage [Uimp]			kV	6	6	6
Rated Thermal Current [Ith] (AC1)			A	45	55	60
AC3	200 ~ 240V		kW/A	5.5/25	7.5/32	11/40
	380 ~ 440V			11/25	15/32	18.5/40
	500 ~ 550V			15/22	18.5/28	22/32
	660 ~ 690V			15/17	18.5/20	22/23
	Lifespan	Electrical	10,000 Times	140	140	140
		Mechanical		1,000	1,000	1,000
AC4	200 ~ 240V		kW/A	3.7/18	4.5/22	5.5/25
	380 ~ 440V			5.5/13	7.5/17	11/24
	Electrical Lifespan		10,000 Times	3	3	3
AC1, 2, 3 Load Permitted Operational Frequency/Time	100% Load		Frequency	1,000	1,000	1,000
	50% Load (DC)			2,000	2,000	2,000
	20% Load (DC)			3,600	3,600	3,600
AC4 Load Permitted Operational Frequency/Time	100% Load		Frequency	250	250	250
	50% Load			500	500	500
Making Capacity	220V		A	250	320	400
	440V			250	320	400
Breaking Capacity	220V		A	200	256	320
	440V			200	256	320
Mounting Method				Screw & Din Rail Mounting		
Standard Auxiliary Contacts				1a1b		
External Dimensions	W×H×D		mm	55×84.6×97.5		
Weight			kg	0.6		
Contact Arrangement						
Standard	Main Contact		3a			
	Auxiliary Contact		1a1b			

Accessories

※ Auxiliary Contact Limitations

Standard + additional maximum number of poles : 8
Maximum number of poles allowed for A contacts : 6
Maximum number of poles allowed for B contacts : 4

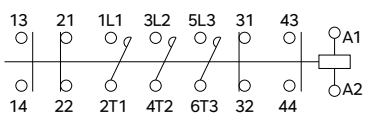


Order Code

Operational Voltage (V)		HGC25B11	HGC25B11	HGC40B11
AC (50Hz)	24	HGC25B 11NS X024	HGC32B 11NS X024	HGC40B 11NS X024
	110	HGC25B 11NS X110	HGC32B 11NS X110	HGC40B 11NS X110
	220	HGC25B 11NS X220	HGC32B 11NS X220	HGC40B 11NS X220
	240	HGC25B 11NS X240	HGC32B 11NS X240	HGC40B 11NS X240
	380	HGC25B 11NS X380	HGC32B 11NS X380	HGC40B 11NS X380
	440	HGC25B 11NS X440	HGC32B 11NS X440	HGC40B 11NS X440
AC (60Hz)	24	HGC25B 11NS A024	HGC32B 11NS A024	HGC40B 11NS A024
	110	HGC25B 11NS A110	HGC32B 11NS A110	HGC40B 11NS A110
	220	HGC25B 11NS A220	HGC32B 11NS A220	HGC40B 11NS A220
	240	HGC25B 11NS A240	HGC32B 11NS A240	HGC40B 11NS A240
	380	HGC25B 11NS A380	HGC32B 11NS A380	HGC40B 11NS A380
	380	HGC25B 11NS A440	HGC32B 11NS A440	HGC40B 11NS A440

Model Selection Table

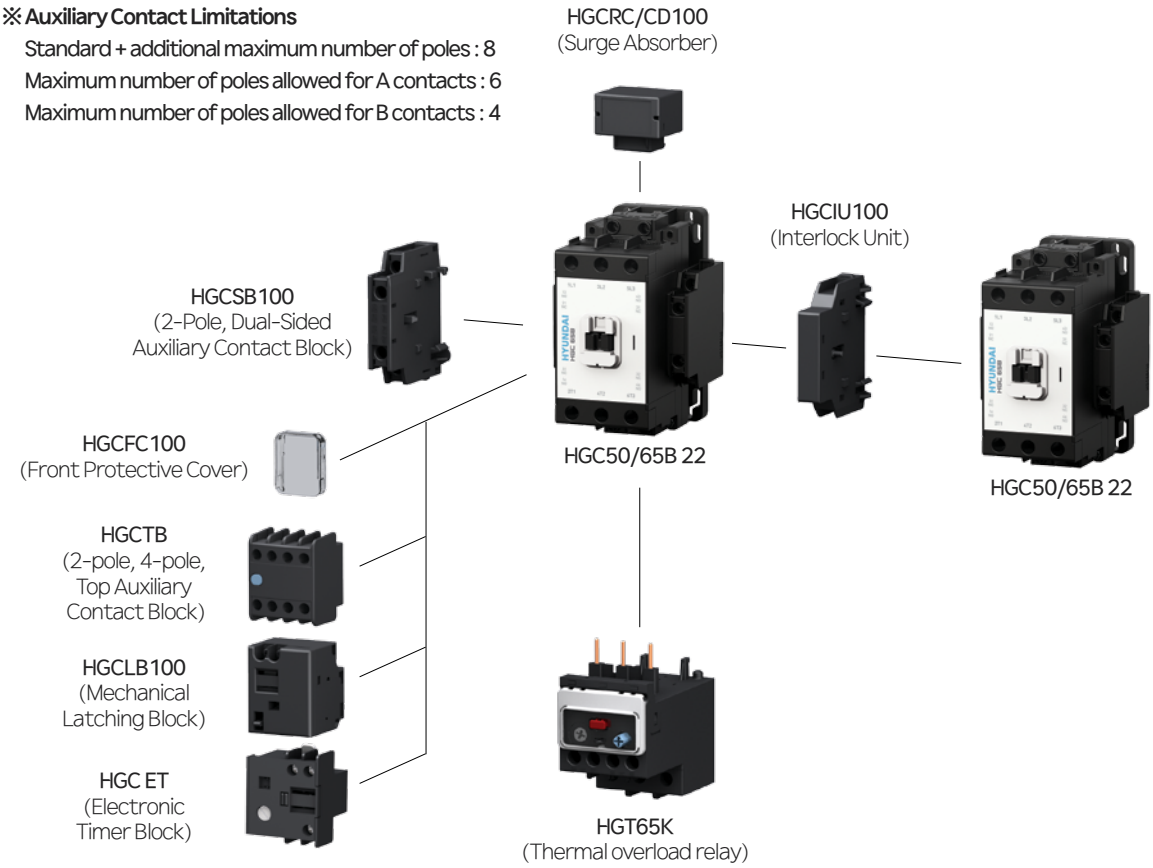
Magnetic Contactor : HGC50/65B 22 (2a2b)

Rating				HGC	
Model				**HGC50B 22**	**HGC65B 22**
IEC60947-4-1					
Pole			P	3	3
Rated Frequency			Hz	50/60	50/60
Rated Insulation Voltage [Ui]			V	1,000	1,000
Rated Operational Voltage [Ue]			V	690	690
Rated Impulse Withstand Voltage [Uimp]			kV	8	8
Rated Thermal Current [Ith] (AC1)			A	70	85
AC3	200 ~ 240V		kW/A	15/50	18.5/65
	380 ~ 440V			22/50	30/65
	500 ~ 550V			30/43	33/60
	660 ~ 690V			30/28	33/35
	Lifespan	Electrical	10,000 Times	200	200
		Mechanical		1,500	1,500
AC4	200 ~ 240V		kW/A	7.5/35	11/50
	380 ~ 440V			15/32	22/47
	Electrical Lifespan		10,000 Times	3	3
AC1, 2, 3 Load Permitted Operational Frequency/Time	100% Load		Frequency	750	750
	50% Load (DC)			1,500 (900)	1,500 (900)
	20% Load (DC)			3,000 (1,200)	3,000 (1,200)
AC4 Load Permitted Operational Frequency/Time	100% Load		Frequency	250	250
	50% Load			500	500
Making Capacity	220V		A	500	650
	440V			500	650
Breaking Capacity	220V		A	400	520
	440V			400	520
Mounting Method				Screw & Din Rail Mounting	
Standard Auxiliary Contacts				2a2b	
External Dimensions	W×H×D		mm	79.8×123.6×107	
Weight			kg	0.85	
Contact Arrangement					
Standard	Main Contact		3a		
	Auxiliary Contact		2a2b		

Accessories

※ Auxiliary Contact Limitations

Standard + additional maximum number of poles : 8
Maximum number of poles allowed for A contacts : 6
Maximum number of poles allowed for B contacts : 4

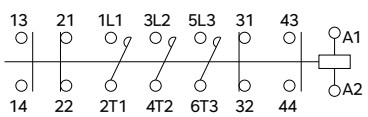


Order Code

Operational Voltage (V)		HGC50B 22	HGC65B 22
AC (50Hz)	24	HGC50B 22NS X024	HGC65B 22NS X024
	110	HGC50B 22NS X110	HGC65B 22NS X110
	220	HGC50B 22NS X220	HGC65B 22NS X220
	240	HGC50B 22NS X240	HGC65B 22NS X240
	380	HGC50B 22NS X380	HGC65B 22NS X380
	440	HGC50B 22NS X440	HGC65B 22NS X440
AC (60Hz)	24	HGC50B 22NS A024	HGC65B 22NS A024
	110	HGC50B 22NS A110	HGC65B 22NS A110
	220	HGC50B 22NS A220	HGC65B 22NS A220
	240	HGC50B 22NS A240	HGC65B 22NS A240
	380	HGC50B 22NS A380	HGC65B 22NS A380
	440	HGC50B 22NS A440	HGC65B 22NS A440

Model Selection Table

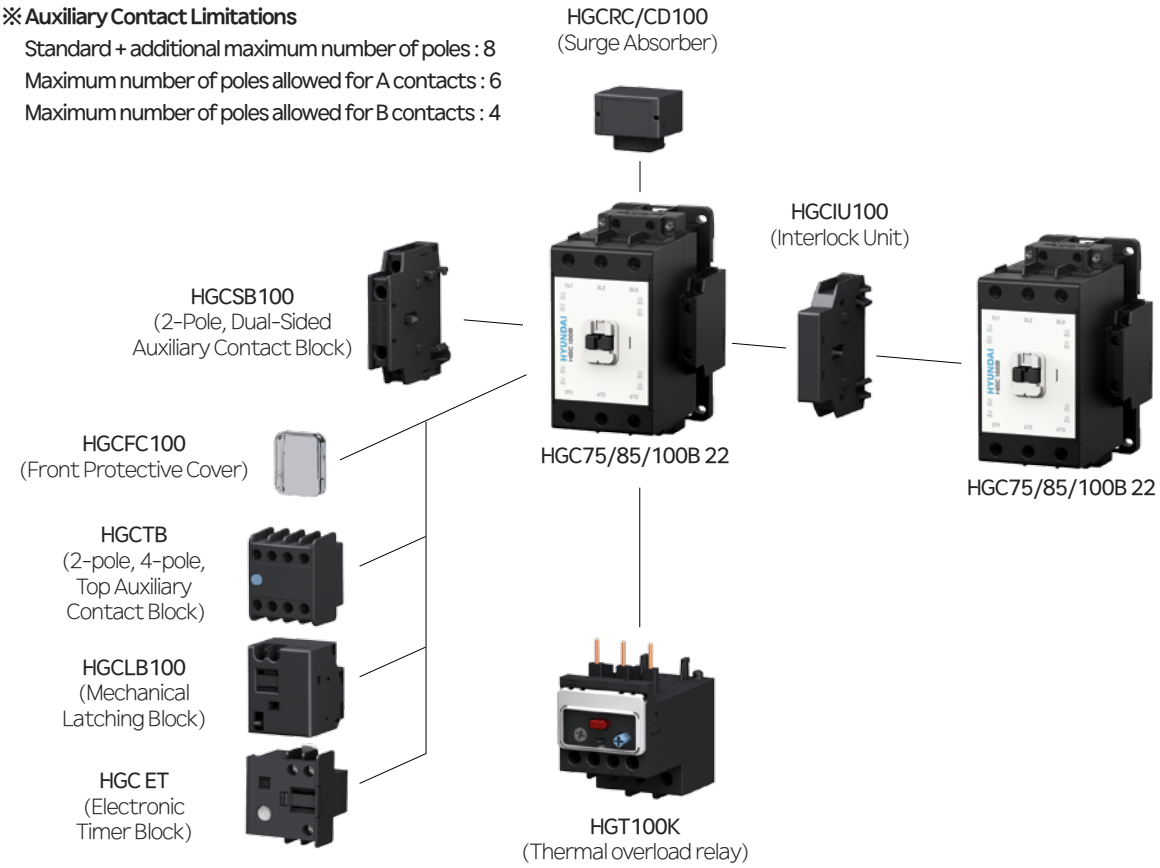
Magnetic Contactor : HGC75/85/100B 22 (2a2b)

Rating				HGC		
Model				**HGC75B22**	**HGC85B22**	**HGC100B22**
IEC60947-4-1						
Pole			P	3	3	3
Rated Frequency			Hz	50/60	50/60	50/60
Rated Insulation Voltage [Ui]			V	1,000	1,000	1,000
Rated Operational Voltage [Ue]			V	690	690	690
Rated Impulse Withstand Voltage [Uimp]			kV	8	8	8
Rated Thermal Current [Ith] (AC1)			A	115	125	135
AC3	200 ~ 240V		kW/A	22/75	25/85	30/100
	380 ~ 440V			37/75	45/85	55/100
	500 ~ 550V			37/64	45/75	55/85
	660 ~ 690V			37/42	45/45	50/65
	Lifespan	Electrical	10,000 Times	200	200	200
		Mechanical		1,000	1,000	1,000
AC4	200 ~ 240V		kW/A	13/55	15/65	17/72
	380 ~ 440V			25/52	30/62	33/68
	Electrical Lifespan		10,000 Times	3	3	3
AC1, 2, 3 Load Permitted Operational Frequency/Time	100% Load		Frequency	450	450	450
	50% Load (DC)			900	900	900
	20% Load (DC)			1,800	1,800	1,800
AC4 Load Permitted Operational Frequency/Time	100% Load		Frequency	200	200	200
	50% Load			400	400	400
Making Capacity	220V		A	750	850	1,000
	440V			750	850	1,000
Breaking Capacity	220V		A	600	680	800
	440V			600	680	800
Mounting Method				Screw & Din Rail Mounting		
Standard Auxiliary Contacts				2a2b		
External Dimensions	W×H×D		mm	94.8×146×130.6		
Weight			kg	1.5		
Contact Arrangement						
Standard	Main Contact		3a			
	Auxiliary Contact		2a2b			

Accessories

※ Auxiliary Contact Limitations

Standard + additional maximum number of poles : 8
Maximum number of poles allowed for A contacts : 6
Maximum number of poles allowed for B contacts : 4



Order Code

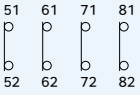
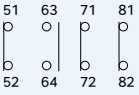
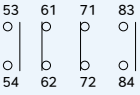
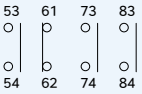
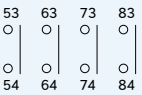
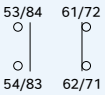
Operational Voltage (V)		HGC75B 22	HGC85B 22	HGC100B 22
AC (50Hz)	24	HGC75B 22NS X024	HGC85B 22NS X024	HGC100B 22NS X024
	110	HGC75B 22NS X110	HGC85B 22NS X110	HGC100B 22NS X110
	220	HGC75B 22NS X220	HGC85B 22NS X220	HGC100B 22NS X220
	240	HGC75B 22NS X240	HGC85B 22NS X240	HGC100B 22NS X240
	380	HGC75B 22NS X380	HGC85B 22NS X380	HGC100B 22NS X380
	440	HGC75B 22NS X440	HGC85B 22NS X440	HGC100B 22NS X440
AC (60Hz)	24	HGC75B 22NS A024	HGC85B 22NS A024	HGC100B 22NS A024
	110	HGC75B 22NS A110	HGC85B 22NS A110	HGC100B 22NS A110
	220	HGC75B 22NS A220	HGC85B 22NS A220	HGC100B 22NS A220
	240	HGC75B 22NS A240	HGC85B 22NS A240	HGC100B 22NS A240
	380	HGC75B 22NS A380	HGC85B 22NS A380	HGC100B 22NS A380
	380	HGC75B 22NS A440	HGC85B 22NS A440	HGC100B 22NS A440

Accessories

Auxiliary Contact Block (TB/SB)

Exterior and Model Name		Rating		
Exterior	Model	Rating of Contact		
Front Mounting 	HGC TB	Rated Insulation Voltage (Ui)	IEC	AC 690V
			UL	AC 600V
		Rated Thermal Current (Ith)		16A
		Minimum load		10mA (24V DC)
		Rated Operational Current AC15 (coil load)	120V	6A
			240V	4A
			380V	3A
			440V	3A
			500V	2A
			690V	2A
		Rated Operational Current DC13 (coil load)	24V	4A
			48V	2.5A
			125V	1.1A
			250V	0.3A
			480V	0.2A
			600V	0.2A

Exterior	Model	Based on IEC60947-5-1			UL and CSA Standards		
Side Mounting 	HGC SB	Rated Insulation Voltage (Ui)		AC750V	Thermal Current		16A
		Rated Thermal Current (Ith)		16A	Rated Operational Current (AC)	120V	6A
		Minimum load		10mA (24V DC)		240V	3A
		Rated Operational Current AC12 (Resistive Load)	110V	10A		480V	1.5A
			220V	8A		600V	1.2A
			440V	6A	Rated Operational Current (DC)	125V	1.1A
			690V	2A		250V	0.55A
		Rated Operational Current AC15 (Coil Load)	110V	6A		440V	0.2A
			220V	4A		600V	0.2A
			440V	3A	※ Contact Rating Code : A600 ~ P300		
			690V	2A			
		Rated Operational Current DC12 (Resistive Load)	24V	4A			
			48V	2.5A			
			125V	1.1A			
			250V	0.3A			
		Rated Operational Current DC13 (Coil Load)	24V	4A			
			48V	2.5A			
			125V	1.1A			
			250V	0.3A			

Contact Arrangement		Order Code	Others		
Combination	Arrangement	With Terminal Cover	Weight	Applicable Magnetic Contactor	Installation Method
2NC		HGC TB02NS	0.031	HGC9B ~ 100B	Front Mounting  ※ The maximum number of b-contacts that can be combined is 4NC.
1NO + 1NC		HGC TB11NS			
2NO		HGC TB20NS			
4NC		HGC TB04NS			
1NO + 3NC		HGC TB13NS	0.053		
2NO + 2NC		HGC TB22NS			
3NO + 1NC		HGC TB31NS			
4NO		HGC TB40NS			
1NO + 1NC		HGC SB40 11NS	0.028	HGC9B ~ 40B	Side Mounting  ※ The maximum number of b-contacts that can be combined is 4NC.
		HGC SB100 11NS	0.053	HGC50B ~ 100B	

Accessories

Interlock Unit (IU)

The interlock unit is an accessory that provides interlock functionality for reversible operation of the contactor and can be configured with a mechanical interlock. Reversible Magnetic contactors with mechanical interlocks are more reliable than electrical interlocks because they are not simultaneously in a closed state.

Handling

- Be sure to use the NC contacts of the magnetic contactor for electrical interlocking in parallel.
- When attaching the product, do not make transverse attachments.
- Do not apply excessive force when manually checking for simultaneous left and right insertion.
  That may cause damage.

Select & Order

Order Code	Applicable Magnetic Contactor	Weight (kg)
HGC IU40	HGC9B ~ 40B	0.03
HGC IU100	HGC50B ~ 100B	0.03



Mechanical Latching Block (LB)

Mechanical latching block mechanically maintains the closed state even when the Operational voltage supplied to the coil of the magnetic contactor and auxiliary relay falls below the rated voltage and the coil is de-energized. The modularized mechanical latching blocks allow for one-touch installation in Magnetic contactors and auxiliary relay.

Ratings and Characteristics

Coil Power Consumption	VA	120
	W	96
Voltage	V	(0.85 ~ 1.1)×Uc
Operational Frequency	Cycle/h	1,200
Operational Voltage	AC	24, 48, 100 ~ 125, 200 ~ 240V 440V
	DC	24, 48, 100 ~ 125, 200 ~ 240V
Mechanical Life	10,000 Times	50
Weight	kg	0.1

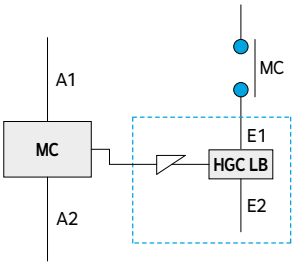
Handling

- The latching block is activated after receiving a mechanical closing signal from the Magnetic contactor or auxiliary relay, and maintains the closed state of the main contact even if the coil of the magnetic contactor or auxiliary relay is de-excited due to a momentary power failure in the line.
- How to Shut off an Magnetic Contactor or Auxiliary Relay
  - Manual operation: Set the button attached to the latching block to the "O" position.
  - Electrical operation: Apply the release voltage to the coil embedded in the latching block.
    Before applying power to the coil in the latching block, the assembled magnetic contactor (relay) must be switched off.

Select & Order

Model	Order Code	Current	Operational Voltage	Applicable Magnetic Contactor
HGC LB100	HGC LB100 F024	AC/DC	24V	HGC9B ~ 100B
	HGC LB100 F048		48V	
	HGC LB100 F110		100 ~ 125V	
	HGC LB100 F220		200 ~ 240V	
	HGC LB100 A440	AC	440V	

Schematic



※ A1/A2 : Coil terminals, E1/E2 : Latching block terminals

- How to Close an Magnetic Contactor or Auxiliary Relay
  - By pressing the button located in the center "I" of the latching block, the coil of the Magnetic contactor or auxiliary relay can be closed without excitation.
- Caution
  - Do not simultaneously energize the Magnetic contactor or auxiliary relay and the latching block.
  - Do not apply power to the power stages (E1, E2) for more than 1 second because the latching block is instantaneous.
  - See the top right schematic for the latching block operation circuit.



Surge Absorber (RC)

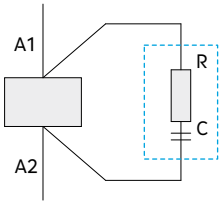
Surge Absorber (RC-Unit)

- Reduce 10 to 20 times higher surge voltage that occurs when a magnetic contactor or auxiliary relays operates to less than about 3 times the rated voltage.
- Ideal for magnetic contactor control circuits using PLCs.
- Protects electronic components from burnout due to high surge voltages.
- Available for 50/60 Hz public use. (For RC-Unit)
- Noise components above 400 Hz and above 5% can cause internal component degradation.

Select & Order

Product	Order Code	Operational Voltage	Applicable Magnetic Contactor	Weight (kg)
RC-Unit	HGC RC40 Y048	AC 24 ~ 48V	HGC9B ~ 40B	0.029
	HGC RC40 Y220	AC 110 ~ 220V		
	HGC RC40 Y380	AC 240 ~ 380V		
	HGC RC100 Y048	AC 24 ~ 48V	HGC50B ~ 100B	
	HGC RC100 Y220	AC 110 ~ 220V		
	HGC RC100 Y380	AC 240 ~ 380V		





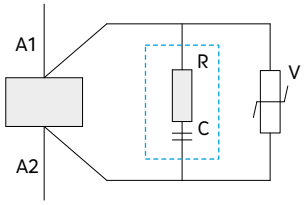
Surge Absorber (Varistor + RC-Unit)

- The HGCRC100 F110/F220 is a varistorized type that can be used for AC/DC applications.
- When a surge voltage above the rated voltage comes in, it is discharged through the varistors to prevent high voltage from entering the device.

Select & Order

Product	Order Code	Operational Voltage	Applicable Magnetic Contactor	Weight (kg)
RC-Unit with Varistor	HGC RC100 F110	AC 48 ~ 110V	HGC9B ~ 100B	0.029
		DC 48 ~ 110V		
	HGC RC100 F220	AC 110 ~ 220V		
		DC 110 ~ 220V		





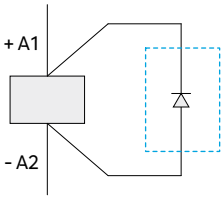
Clamping Diode

- Used to prevent reverse electromotive force on the DC power stage.
- If Coils are connected in parallel, be sure to attach them.
- Prevent reverse electromotive force and electronic components from damage by a fire.

Select & Order

Product	Order Code	Operational Voltage	Applicable Magnetic Contactor	Weight (kg)
Clamping Diode	HGC CD100	DC 24 ~ 220V	HGC9B ~ 100B	0.029





Accessories

Electronic Timer Block (ET)

- The modular design is one touch attachment to magnetic contacts and auxiliary relays which allows to save spce.
- Designed for a wide range of Operational voltages and AC/DC dual use.
- A single timer can be implemented for both On-Delay and Interval timer functionality and is optimal as the Interval timer is for Y-Δ starters.

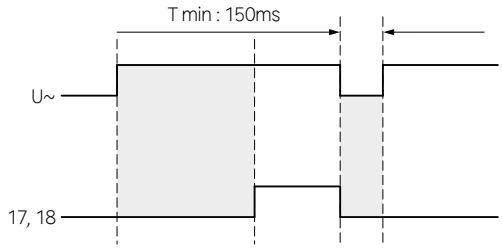
Select & Order

Order Code	Applicable MagneticContactor	Operational Voltage
HGC ET1	HGC9B ~ 100B	AC/DC 90 ~ 240V
HGC ET2		AC/DC 24 ~ 60V

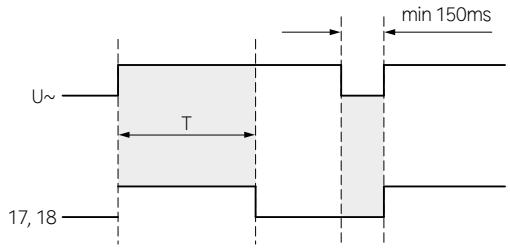
Ratings and Characteristics

Model	Order Code	HGC ET1	HGC ET2
Rated Voltage	AC/DC V	90 ~ 240	24 ~ 60
Allowable Voltage	V	(0.8 ~ 1.1)×Rated Voltage	
Decommissioning Capacity	VA	90	
Maximum Load	VA	15	
Delay Time	Position A	10 ~ 220	
	Position B	0.15 ~ 15	
Accuracy	%	±5	
Repeatability Error	%	0.1	
	Secs	50ms	
Weight	kg	0.053	

Operational Concept Map



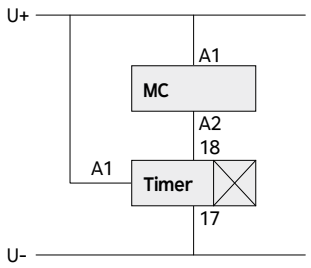
• Location 1 : On-Delay / t : Adjustable Time 0.15....220sec



• Location 2 : Interval Timer

※ Requires 150 ms pause time for reactivation after power off





※ A1/A2 : Coil Terminals, 17/18 : Timer terminals

Manipulator Coil HGC9B ~ 100B

※ 5 pieces of manuipulator coils as a package in a single part number.

Operational Voltage		Order Code				
		Applicable Magnetic Contactor				
Frequency	V	HGC9 ~ 22B10 HGC9 ~ 22B01	HGC9 ~ 22B11	HGC22 ~ 40B11	HGC50 ~ 65B22	HGC75 ~ 100B22
AC 60Hz	24	HGCOLPKG18BA024	HGCOLPKG22BA024	HGCOLPKG40BA024	HGCOLPKG65BA024	HGCOLPKG100BA024
	110	HGCOLPKG18BA110	HGCOLPKG22BA110	HGCOLPKG40BA110	HGCOLPKG65BA110	HGCOLPKG100BA110
	120	HGCOLPKG18BA120	HGCOLPKG22BA120	HGCOLPKG40BA120	HGCOLPKG65BA120	HGCOLPKG100BA120
	220	HGCOLPKG18BA220	HGCOLPKG22BA220	HGCOLPKG40BA220	HGCOLPKG65BA220	HGCOLPKG100BA220
	240	HGCOLPKG18BA240	HGCOLPKG22BA240	HGCOLPKG40BA240	HGCOLPKG65BA240	HGCOLPKG100BA240
	440	HGCOLPKG18BA440	HGCOLPKG22BA440	HGCOLPKG40BA440	HGCOLPKG65BA440	HGCOLPKG100BA440
AC 50Hz	24	HGCOLPKG18BX024	HGCOLPKG22BX024	HGCOLPKG40BX024	HGCOLPKG65BX024	HGCOLPKG100BX024
	48	HGCOLPKG18BX048	HGCOLPKG22BX048	HGCOLPKG40BX048	HGCOLPKG65BX048	HGCOLPKG100BX048
	110	HGCOLPKG18BX110	HGCOLPKG22BX110	HGCOLPKG40BX110	HGCOLPKG65BX110	HGCOLPKG100BX110
	220	HGCOLPKG18BX220	HGCOLPKG22BX220	HGCOLPKG40BX220	HGCOLPKG65BX220	HGCOLPKG100BX220
	230	HGCOLPKG18BX230	HGCOLPKG22BX230	HGCOLPKG40BX230	HGCOLPKG65BX230	HGCOLPKG100BX230
	240	HGCOLPKG18BX240	HGCOLPKG22BX240	HGCOLPKG40BX240	HGCOLPKG65BX240	HGCOLPKG100BX240
	380	HGCOLPKG18BX380	HGCOLPKG22BX380	HGCOLPKG40BX380	HGCOLPKG65BX380	HGCOLPKG100BX380
	440	HGCOLPKG18BX440	HGCOLPKG22BX440	HGCOLPKG40BX440	HGCOLPKG65BX440	HGCOLPKG100BX440

Contact & Cover

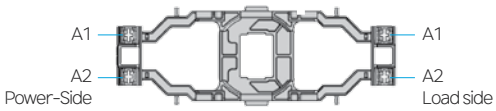
Exterior	Model	Applicable Product	Order Code	When Ordering 1 Unit (Set)
Terminal Cover  Main Terminal Cover  Coil Terminal Cover	HGCPC	HGC9 ~ 22B10/01	HGCPC22B10NS	5
		HGC9 ~ 22B11	HGCPC22B11NS	
		HGC25 ~ 40B11	HGCPC40B	
		HGC50 ~ 65B22	HGCPC65B	
		HGC75 ~ 100B22	HGCPC100B	
Front Cover 	HGCFC	HGC9 ~ 100B	HGCFC100	4

Technical Data

AC Control Coil Selection Table

Magnetic Contactor Model Name			HGC9/12/18/22B 10 HGC9/12/18/22B 01	HGC9/12/18/22B 11	HGC25/32/40B 11	HGC50/65B 22	HGC75/85/100B 22
Rated Operational Voltage		[Uc]					
	50Hz	[V]	24, 48, 110, 220, 230, 380, 440				
	60Hz	[V]	24, 110, 120, 220, 240, 440				
Operational Voltage Range			85 ~ 110%				
Coil Power Consumption(20℃, Rated Voltage)							
AC 220V 60Hz	Input	VA	80	80	80	200	300
	Retention	VA	10	10	10	15	25
Action Time	Input	ms	8 ~ 30	8 ~ 30	8 ~ 30	8 ~ 30	80 ~ 30
	Released		10 ~ 30	10 ~ 30	10 ~ 30	10 ~ 30	10 ~ 30

※ 1. Coil power consumption is similar across voltage ratings, but there may be some differences.
2. For HGC9 ~ 22B10/01, HGC65B, and HGC100B, A1 and A2 terminals on the power side and load side are connected, so be careful not to cause a short circuit when wiring.



Auxiliary Contact Selection Table

IEC 60947

Rated Insulation Voltage (Ui)	AC 750V							
Rated Operational Current (Ith)	16A							
Minimum load	10mA (24V DC)							
Rated Current	AC12 (Resistive Loads)		AC15 (Coil Load)		DC12 (Resistive Load)		DC13 (Coil Load)	
	110V	10A	110V	6A	24V	4A	24V	4A
	220V	8A	220V	4A	48V	2.5A	48V	2.5A
	440V	6A	440V	3A	125V	1.1A	125V	1.1A
	690V	2A	690V	2A	220V	0.3A	220V	0.3A

UL & CSA

Current	16A			
Minimum load	10mA (24V DC)			
Rated Current	AC		DC	
	120V	6A	125V	1.1A
	240V	3A	250V	0.3A
	480V	1.5A	440V	0.2A
	600V	1.2A	600V	0.2A

※ Contact Rating Code : A600 ~ P150

Accessible Wire & Tightening Torque

Type	Main Terminal Size	Stranded Wire + Pinterminal (mm^2)		Solid Wire (mm^2)	Stranded Wire (mm^2)	Maximum Terminal Width (mm)	Tightening Torque [kgf.cm]
HGC9/12/18/22B 10/01	M3.5	1…10	1…6	1…10	1…10	8	12
HGC9/12/18/22B 11	M4	1…10	1…6	1…10	1…10	10	15
HGC25/32/40B	M5	1…10	1…6	2.5…16	2.5…16	12	26
HGC50/65B	M6	N/A		6…25	6…25	13	40
HGC75/85/100B	M8	N/A		10…50	10…50	17	60
Aux, Coil	M3.5	0.5…5.5		0.75…5.5	0.75…5.5	7.5	12

Order Code

HGC

HGC	12	22	N	S	A	220
Model	Motor Rated Current	Composition of Auxiliary Contacts	Auxiliary Contacts Usage	Protective Cover	Operational Coil Frequency	Operational Coil Voltage

Model

HGC	Magnetic Contactor

Motor Rated Current

Code	Rated Current	Capacity
	AC3/AC 440V	
9B	9A	4kW
12B	12A	5.5kW
18B	18A	7.5kW
22B	22A	11kW
25B	25A	11kW
32B	32A	15kW
40B	40A	18.5kW
50B	50A	22kW
65B	65A	30kW
75B	75A	37kW
85B	85A	45kW
100B	100A	55kW

Composition of Auxiliary Contacts

10	1NO	HGC9 ~ 22B10
01	1NC	HGC9 ~ 22B01
11	1NO + 1NC	HGC9 ~ 40B11
22	2NO + 2NC	HGC50B ~ 100B

Auxiliary Contacts Usage

N	General type

Protective Cover

S	Protective cover is attached on main contact, auxiliary contact and coil

Operational Coil Frequency

X	AC 50Hz
A	AC 60Hz

Operational Coil Voltage

24, 48, 110, 220, 230, 240, 380, 440
24, 110, 120, 220, 240, 440

HG Series

Thermal Overload Relays

Provides protection solutions to safeguard electric motors in various circumstances with fast and accurate overload TRIP performance



TOR Safety Protective Cover

Protective covers are applied as a standard feature to all products to ensure safe use

Separate Test/Reset buttons

If an emergency stop is required while the motor is running,

the Test button allows for a quick and accurate stop

Quick trials are possible with Test button when testing contact behavior



Magnetic Contactor
(HGC B Type)



Overload Relay
(HGT B Type)

Model Selection Table

Thermal Overload Relay (HGT)

Applicable Standards	IEC/EN 60947-4-1
Acquired Certificates	CE, KC
Rated Operational Voltage	Max. 690V
Rated Insulation Voltage	690V
Rated Frequency	50/60Hz

Exterior and Model Name		Rating							Order code		
		Current [A]				Applicable Cable(mm^2)			Protection Grade 10 A		
Exterior	Model	Nominal Rating	Setting Current			Main Circuit	Aux. Circuit	Aux. Contact	Applicable Magnetic Contactor	Screw Type Terminal (With Terminal Cover)		Accessory Independent Installation Unit
			Min.	Med.	Max.					K-Type 3-Pole, 3 Elements	H-Type 2-Pole, 2 Elements	
 External Dimension (mm) : 45×82.7×78.2 (W×H×D) Weight (kg) : 0.12	HGT18KB	0.18	0.12	0.15	0.18	1 ~ 2.5	1 ~ 1.25	1NO 1NC	HGC9B 10 HGC12B 10 HGC18B 10 HGC22B 10 HGC9B 01 HGC12B 01 HGC18B 01 HGC22B 01	HGT18KB A0P18S	NA	N/A
		0.26	0.18	0.22	0.26					HGT18KB A0P226S	NA	
		0.35	0.25	0.3	0.35					HGT18KB A0P35S	NA	
		0.5	0.34	0.42	0.5					HGT18KB A0P50S	NA	
		0.7	0.5	0.6	0.7					HGT18KB A0P70S	NA	
		0.9	0.6	0.75	0.9					HGT18KB A0P90S	NA	
		1.2	0.8	1	1.2					HGT18KB A1P20S	NA	
		1.6	1.1	1.35	1.6					HGT18KB A1P60S	NA	
		2.1	1.5	1.8	2.1					HGT18KB A2P10S	NA	
		3	2	2.5	3					HGT18KB A0003S	NA	
		4.2	2.8	3.5	4.2					HGT18KB A4P20S	NA	
		5	3	4	5					HGT18KB A0005S	NA	
		6	4	5	6					HGT18KB A0006S	NA	
		8	5.6	6.8	8					HGT18KB A0008S	NA	
		9	6	7.5	9					HGT18KB A0009S	NA	
		12	8	10	12					HGT18KB A0012S	NA	
		18	12	15	18					HGT18KB A0018S	NA	
		22	15	18.5	22	2 ~ 10				HGT18KB A0022S	NA	

Exterior and Model Name		Rating							Order code		
		Current [A]				Applicable Cable(mm^2)			Protection Grade 10 A		
Exterior	Model	Nominal Rating	Setting Current			Main Circuit	Aux. Circuit	Aux. Contact	Applicable Magnetic Contactor	Screw Type Terminal (With Terminal Cover)		Accessory Independent Installation Unit
			Min.	Med.	Max.					K-Type 3-Pole, 3 Elements	H-Type 2-Pole, 2 Elements	
 External Dimension (mm) : 45×82.7×78.2 (W×H×D) Weight (kg) : 0.12	HGT22KC	0.18	0.12	0.15	0.18	1 ~ 2.5	1 ~ 1.25	1NO 1NC	HGC9B 11 HGC12B 11 HGC18B 11 HGC22B 11	HGT22KC A0P18S	NA	N/A
		0.26	0.18	0.22	0.26					HGT22KC A0P226S	NA	
		0.35	0.25	0.3	0.35					HGT22KC A0P35S	NA	
		0.5	0.34	0.42	0.5					HGT22KC A0P50S	NA	
		0.7	0.5	0.6	0.7					HGT22KC A0P70S	NA	
		0.9	0.6	0.75	0.9					HGT22KC A0P90S	NA	
		1.2	0.8	1	1.2					HGT22KC A1P20S	NA	
		1.6	1.1	1.35	1.6					HGT22KC A1P60S	NA	
		2.1	1.5	1.8	2.1					HGT22KC A2P10S	NA	
		3	2	2.5	3					HGT22KC A0003S	NA	
		4.2	2.8	3.5	4.2					HGT22KC A4P20S	NA	
		5	3	4	5					HGT22KC A0005S	NA	
		6	4	5	6					HGT22KC A0006S	NA	
		8	5.6	6.8	8					HGT22KC A0008S	NA	
		9	6	7.5	9					HGT22KC A0009S	NA	
		12	8	10	12					HGT22KC A0012S	NA	
		18	12	15	18					HGT22KC A0018S	NA	
		22	15	18.5	22	2 ~ 10				HGT22KC A0022S	NA	
 External Dimension (mm) : 45×76×68 (W×H×D) Weight (kg) : 0.12	HGT18KD	0.16	0.1	0.12, 0.14	0.16	1~2.5	1~1.25	1NO 1NC	HGC9B 10 HGC12B 10 HGC18B 10	HGT18KD A0P16_S	NA	N/A
		0.25	0.16	0.19, 0.22	0.25					HGT18KD A0P25_S	NA	
		0.4	0.25	0.3, 0.35	0.4					HGT18KD A0P40_S	NA	
		0.63	0.4	0.5	0.63					HGT18KD A0P63_S	NA	
		1	0.63	0.8	1					HGT18KD A0001_S	NA	
		1.6	1	1.2, 1.4	1.6				HGC9B 01 HGC12B 01 HGC18B 01	HGT18KD A1P60_S	NA	
		2.5	1.6	1.9, 2.2	2.5					HGT18KD A2P50_S	NA	
		4	2.5	3, 3.5	4					HGT18KD A0004_S	NA	
		6	4	5	6				HGC9B 11 HGC12B 11 HGC18B 11	HGT18KD A0006_S	NA	
		8	5.5	7	8					HGT18KD A0008_S	NA	
		10	7	8.5	10					HGT18KD A0010_S	NA	
		13	9	11	13					HGT18KD A0013_S	NA	
		18	12	15	18					HGT18KD A0018_S	NA	

Model Selection Table

Thermal Overload Relay (HGT)

Applicable Standards	IEC/EN 60947-4-1
Acquired Certificates	CE, KC
Rated Operational Voltage	Max. 690V
Rated Insulation Voltage	690V
Rated Frequency	50/60Hz

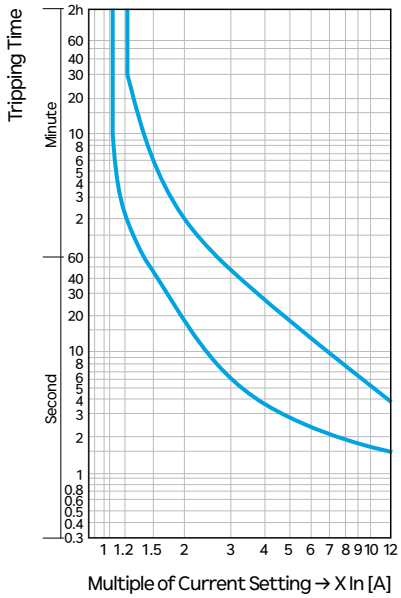
Exterior and Model Name		Rating							Order code		
		Current [A]				Applicable Cable(mm^2)			Protection Grade 10 A		
			Setting Current						Screw Type Terminal (With Terminal Cover)		Accessory Independent Installation Unit
Exterior	Model	Nominal Rating	Min.	Med.	Max.	Main Circuit	Aux. Circuit	Aux. Contact	Applicable Magnetic Contactor	K-Type 3-Pole, 3 Elements	H-Type 2-Pole, 2 Elements	
 External Dimension (mm) : 45×95.5×69.4 (W×H×D) Weight (kg) : 0.16	HGT40KB	10	7	8.5	10	2 ~ 10	1 ~ 2.5	1NO 1NC	HGC25B 11 HGC32B 11 HGC40B 11	HGT40KB A0010S	NA	N/A
		12	8	10	12					HGT40KB A0012S	NA	
		18	12	15	18					HGT40KB A0018S	NA	
		22	15	18.5	22					HGT40KB A0022S	NA	
		25	17	21	25					HGT40KB A0025S	NA	
		32	22	27	32					HGT40KB A0032S	NA	
		40	28	34	40					HGT40KB A0040S	NA	
 External Dimension (mm) : 55×110.7×89.3 (W×H×D) Weight (kg) : 0.29	HGT65	10	7	8.5	10	2 ~ 25	1 ~ 2.5		HGC50B 22 HGC65B 22	HGT65K A0010S	HGT65H A0010S	HGTMB65
		12	8	10	12					HGT65K A0012S	HGT65H A0012S	
		18	12	15	18					HGT65K A0018S	HGT65H A0018S	
		22	15	18.5	22					HGT65K A0022S	HGT65H A0022S	
		25	17	21	25					HGT65K A0025S	HGT65H A0025S	
		32	22	27	32					HGT65K A0032S	HGT65H A0032S	
		40	28	34	40					HGT65K A0040S	HGT65H A0040S	
		50	34	42	50					HGT65K A0050S	HGT65H A0050S	
		65	45	55	65					HGT65K A0065S	HGT65H A0065S	
 External Dimension (mm) : 70×128.1×105 (W×H×D) Weight (kg) : 0.47	HGT100	25	17	21	25	6 ~ 38	1 ~ 2.5		HGC75B 22 HGC85B 22 HGC100B 22	HGT100K A0025S	HGT100H A0025S	HGTMB100
		32	22	27	32					HGT100K A0032S	HGT100H A0032S	
		40	28	34	40					HGT100K A0040S	HGT100H A0040S	
		50	34	42	50					HGT100K A0050S	HGT100H A0050S	
		65	45	55	65					HGT100K A0065S	HGT100H A0065S	
		75	52	63	75					HGT100K A0075S	HGT100H A0075S	
		85	59	72	85					HGT100K A0085S	HGT100H A0085S	
		100	70	85	100					HGT100K A0100S	HGT100H A0100S	

Technical Data

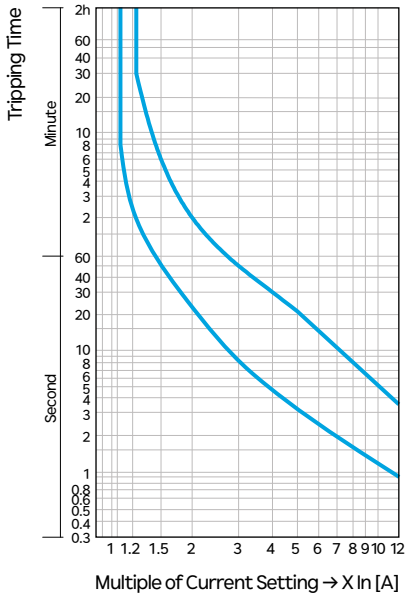
Characteristic Curve

- The trip curves for three-phase overloads show the average trip times based on the phase of cold start with ambient temp. 20
- The operational range of hot start is 20~40% of cold start, so there may be a difference in operating characteristic curve.
- The trip curve for overloaded phase sequence has the 40~60% of trip time for a three-phase overload.

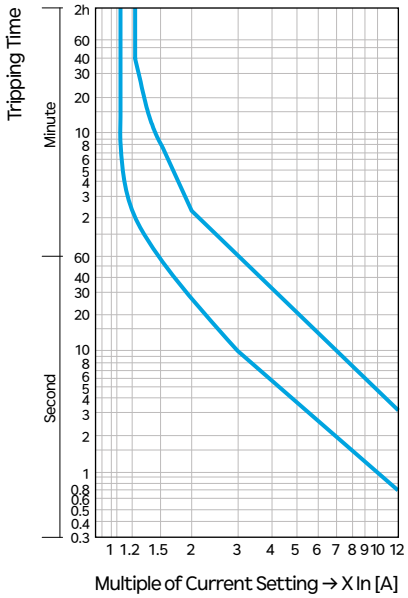
HGT18K



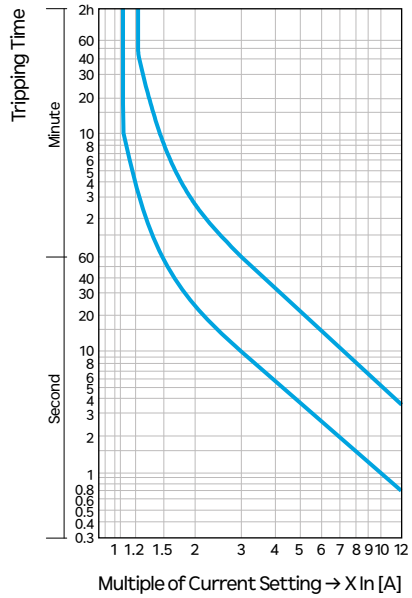
HGT40K



HGT65K



HGT100K



Technical Data

Closing and Breaking Currents of Auxiliary Contacts

Classification	AC15 [1]	
Voltage (V)	Auxiliary Contact 95 ~ 96	Auxiliary Contact 97 ~ 98
	Ie (A)	Ie (A)
110	2	1.2
220	1.5	1
500	1	0.5
660	0.5	0.3

Classification	DC13 [2]	
Voltage (V)	Auxiliary Contact 95 ~ 96	Auxiliary Contact 97 ~ 98
	Ie (A)	Ie (A)
24	1	1
110	0.4	0.4
220	0.15	0.15
440	0.07	0.07

※ 1) AC15 : Closing and Breaking Currents = Ie×10
2) DC13 : Closing and Breaking Currents = Ie×1.1

Order Code

HGT

HGT	18KB	A	0018	S

Model

HGT	Thermal Overload Relay

Capacity of Contactor

18KB	HGC9 ~ 22B10/01
18KD	HGC9~18B10/01 HGC9~18B11
22KC	HGC9 ~ 22B11
40K	HGC25 ~ 40B
65K	HGC50 ~ 65B
100K	HGC75 ~ 100B

Protection Grade (Characteristic Curve)

A	10A

Setting Current

18KB 22KC	0P18	0.12 ~ 0.18A
	0P26	0.18 ~ 0.26A
	0P35	0.25 ~ 0.35A
	0P50	0.34 ~ 0.5A
	0P70	0.5 ~ 0.7A
	0P90	0.6 ~ 0.9A
	1P20	0.8 ~ 1.2A
	1P60	1.1 ~ 1.6A
	2P10	1.5 ~ 2.1A
	0003	2 ~ 3A
	4P20	2.8 ~ 4.2A
	0005	3 ~ 5A
	0006	4 ~ 6A
	0008	5.6 ~ 8A
	0009	6 ~ 9A
	0012	8 ~ 12A
	0018	12 ~ 18A
	0022	15 ~ 22A
18KD	0P16	0.1~0.16
	0P25	0.16~0.25
	0P40	0.25~0.4
	0001	0.63~1
	1P60	1~1.6
	2P50	1.6~2.5
	0004	2.5~4
	0006	4~6
	0008	5.5~8
	0010	7~10
	0013	9~13
	0018	12~18
40K	0010	7 ~ 10A
	0012	8 ~ 12A
	0018	12 ~ 18A
	0022	15 ~ 22A
	0025	17 ~ 25A
	0032	22 ~ 32A
	0040	28 ~ 40A
65K	0010	7 ~ 10A
	0012	8 ~ 12A
	0018	12 ~ 18A
	0022	15 ~ 22A
	0025	17 ~ 25A
	0032	22 ~ 32A
	0040	28 ~ 40A
	0050	34 ~ 50A
	0065	45 ~ 65A
100K	0025	17 ~ 25A
	0032	22 ~ 32A
	0040	28 ~ 40A
	0050	34 ~ 50A
	0065	45 ~ 95A
	0075	52 ~ 75A
	0085	59 ~ 85A
	0100	70 ~ 100A

Protective Cover

S	Protective cover is attached on main contact, auxiliary contact

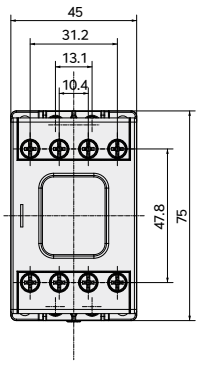
Dimensions

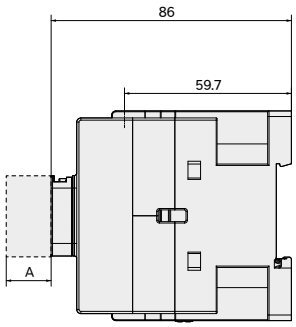
HGC9/12/18/22B 10/01

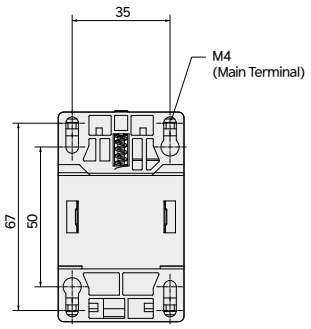
Unit : mm

HGC9
HGC12
HGC18
HGC22B10
HGC22B01

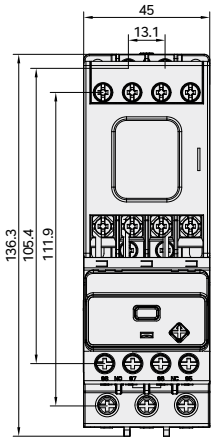
Accessory	A (mm)
Auxiliary Contact HGCTB	35
Latching Blocks HGCLB	42.5
Timer HGCET	39

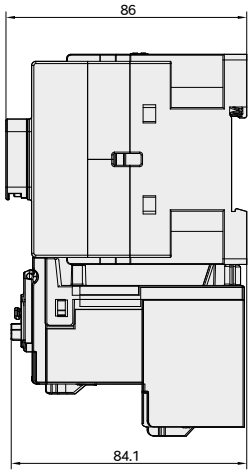


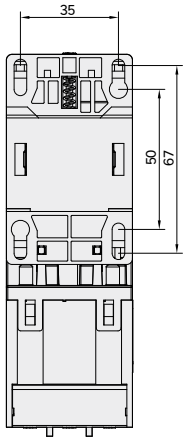




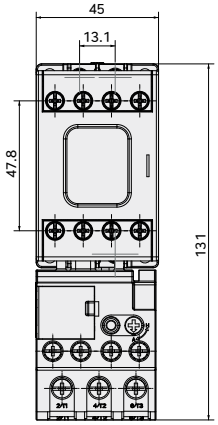
HGC22B +
HGT18KB

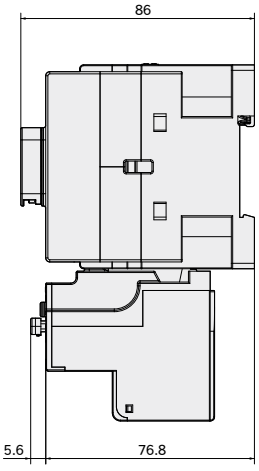


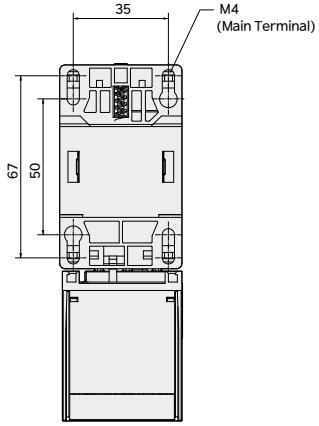




HGC18B10/01
+ HGT18KD







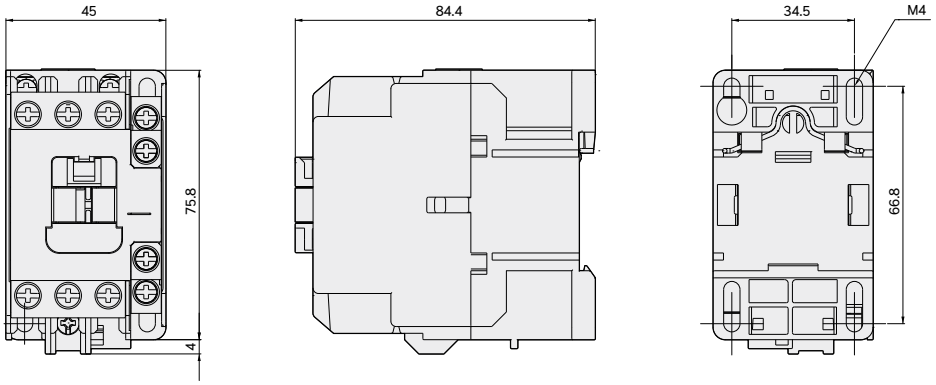
※ The drawing dimensions on this page may change without prior notice.

HGC9/12/18/22B 11

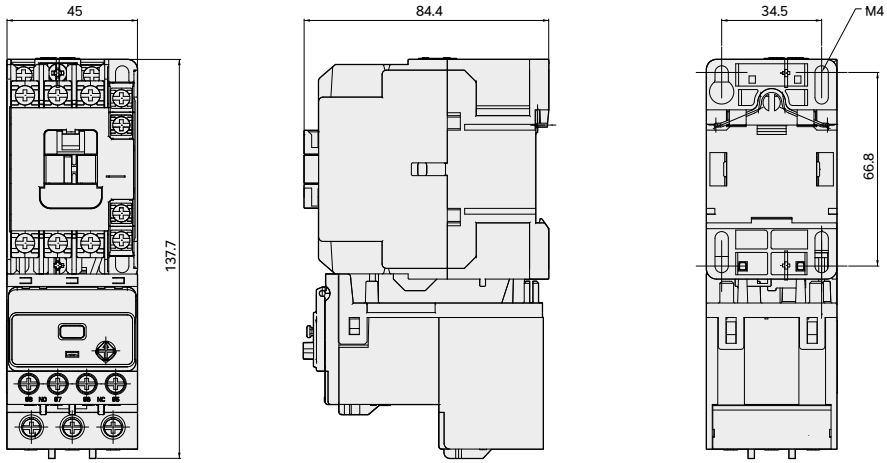
Unit : mm

HGC9
HGC12
HGC18
HGC22B 11

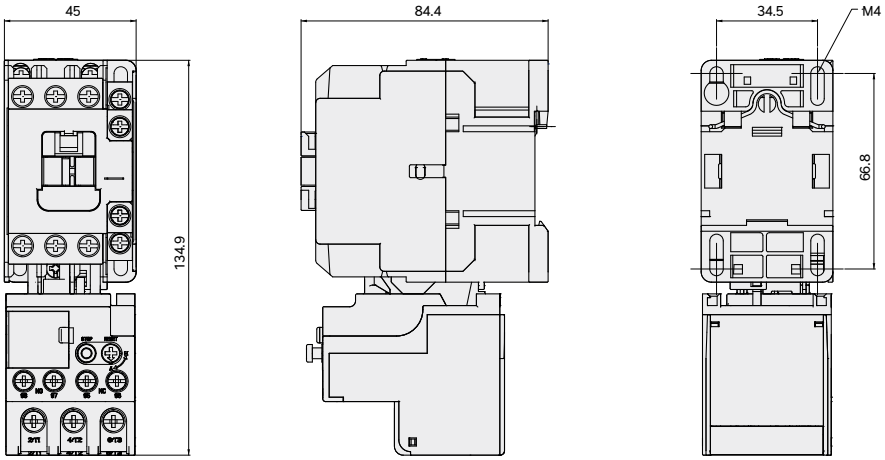
Accessory	A (mm)
Auxiliary Contact HGCTB	35
Latching Blocks HGCLB	42.5
Timer HGCET	39



HGC22B 11
+ HGT22KB



HGC18B11
+ HGT18KD



※ The drawing dimensions on this page may change without prior notice.

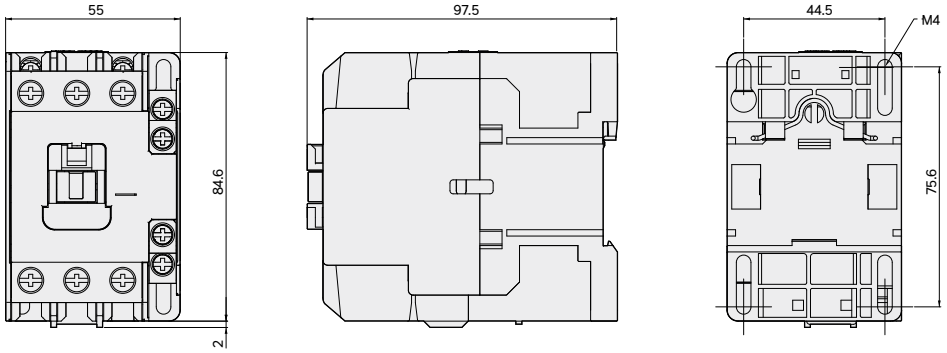
Dimensions

HGC25/32/40B 11

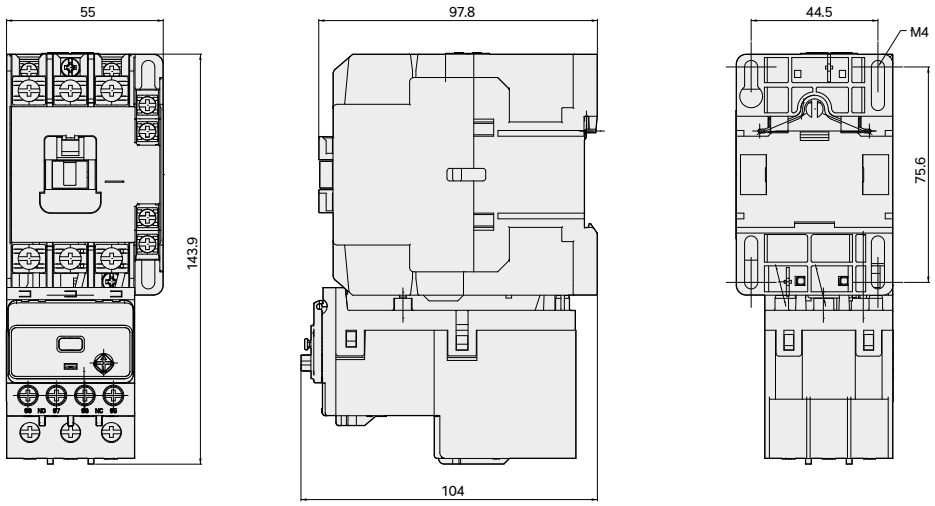
Unit : mm

HGC25
HGC32
HGC40B 11

Accessory	A (mm)
Auxiliary Contact HGCTB	35
Latching Blocks HGCLB	42.5
Timer HGCET	39



HGC25
HGC32
HGC40B 11
+ HGT40KB



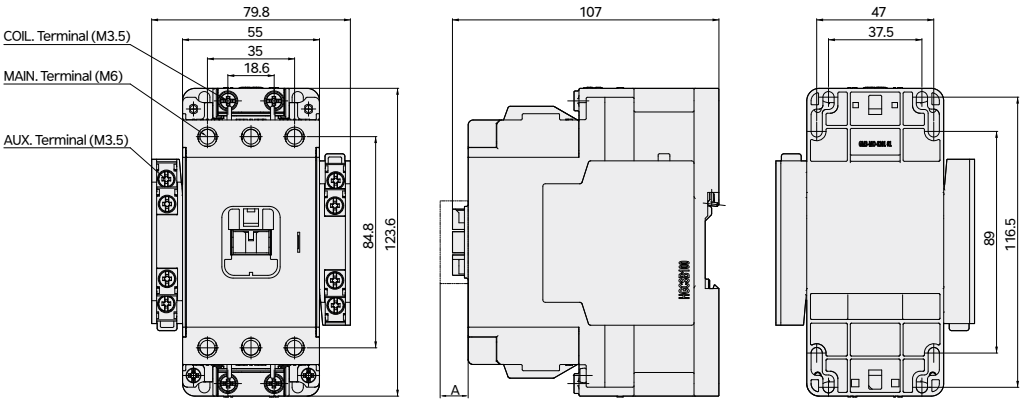
※ The drawing dimensions on this page may change without prior notice.

HGC50/65B 22

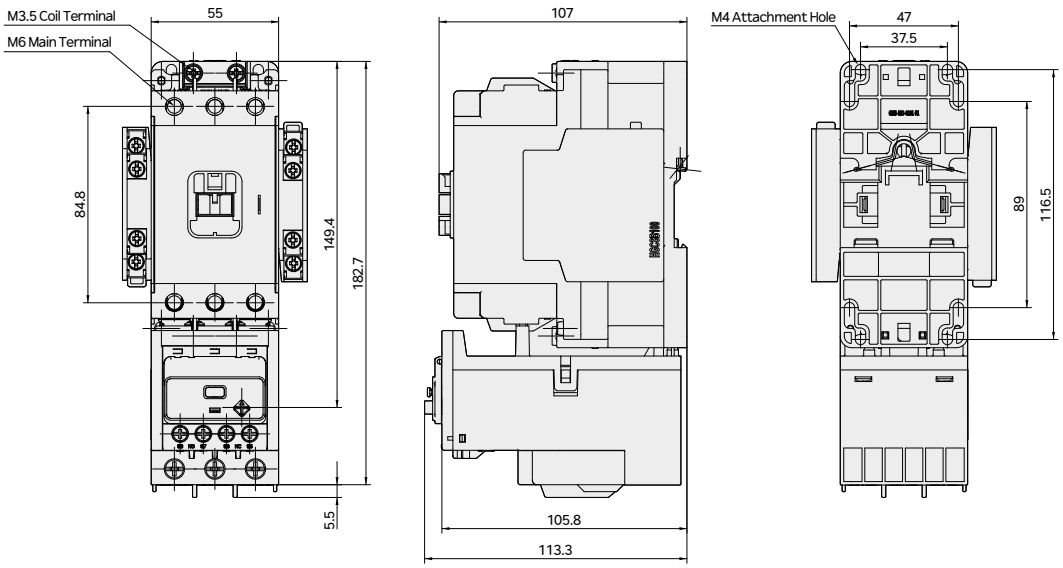
Unit : mm

HGC50B
HGC65B 22

Accessory	A (mm)
Auxiliary Contact HGCTB	35
Latching Blocks HGCLB	42.5
Timer HGCET	39



HGC50B
HGC65B 22
+ HGT65K



※ The drawing dimensions on this page may change without prior notice.

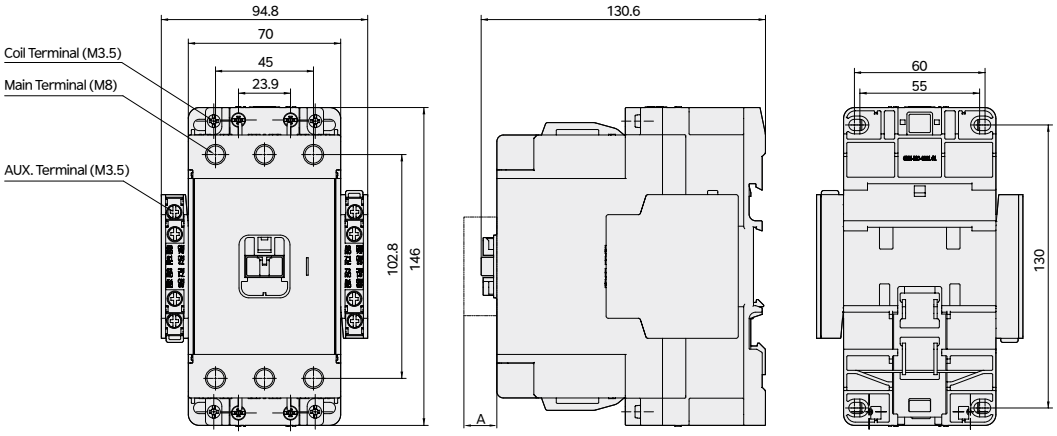
Dimensions

HGC75/85/100B 22

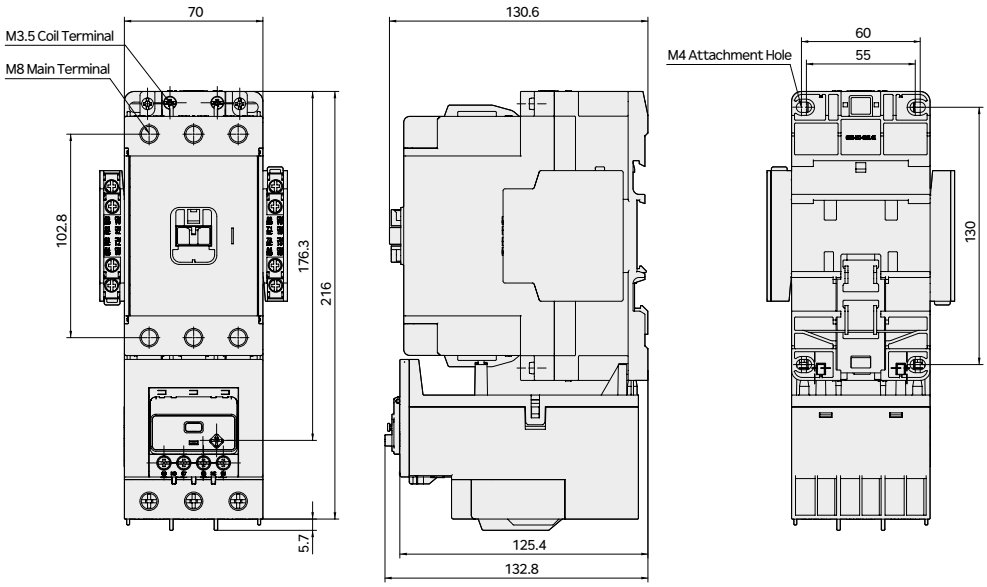
Unit : mm

HGC75B
HGC85B
HGC100B 22

Accessory	A (mm)
Auxiliary Contact HGCTB	35
Latching Blocks HGCLB	42.5
Timer HGCET	39



HGC75
HGC85
HGC100 22
+ HGT100K



※ The drawing dimensions on this page may change without prior notice.

Handling and Maintenance Inspection

Storage and Transportation

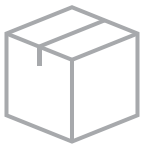
Storage Precaution

| Ambient Temperature | -5℃~+40℃
(Average Temp. 35℃ or less)
-40℃~+50℃ (HGC products only)
| Storage Temperature | -20℃ ~ +60℃
| Altitude | Below 2,000m above sea level
| Relative Humidity | Within 45% to 85%

The surrounding environment may affect the insulation performance and durability of the magnetic contactor so the environment condition for usage must accurately be checked before application.



- **Relative humidity to be within the range of 45% ~ 85%**
  Do not store in places with high humidity for a long period of time.
- **Avoid places with a lot of dust**
  Do not store in exposed places and use cover or packing material to prevent dust from piling up on the circuit breaker.



- **Store in packaged state**
  During storage, store in packaged state. Store in a state using shelf or equivalent devices and do not leave it neglected on the floor. Do not transport heavy products manually. It may cause injury.



- **Do not leave under direct sunlight for a long period of time.**



- **Avoid storage in high or low temperature**
  Storage temperature must be maintained between - 20℃ ~ + 60℃.



- **Do not store in places with corrosive gas**
  Do not leave it around gas containing sulfurous gas or sulfur or ammonia gas and others.

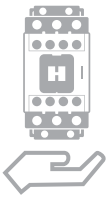


In case of mountainous region with high altitude, it may drop the insulation durability so check the adjustment coefficient of the insulation performance and use the appropriate ratings of the product.

Transportation Precautions

⚠ Caution

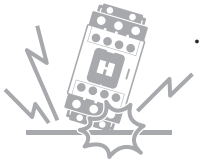
- Do not apply impact during transportation. Dropping or applying strong impact may cause defect.
- Do not handle while holding the circuit breaker's accessory or external plug-in wire of the accessory. It may cause injury in the handler or a malfunction of the circuit breaker.



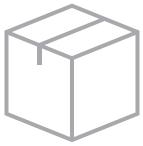
- **Hold the main unit of the circuit breaker during transportation**
  Do not handle while holding the external guide line of the accessory or the terminal bar.



- **Pay attention when handling metal accessories**
  Sharp planes or edges in the metal accessory may cause injury.



- **Do not apply impact during transportation**
  Dropping or applying strong impact may cause defect.



- **Pay attention to the packaging of the circuit breaker before transportation**
  Inappropriate packaging may cause damage in the circuit breaker during transportation.

Handling and Maintenance Inspection

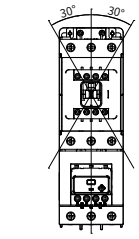
Storage and Transportation

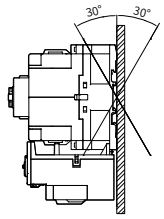
Installation

- Mount in dry places with less vibration.
- As for the installation direction, vertical installation is recommended but it can be used in each direction as seen in the figure.
- In case of cross installation and horizontal installation, it may drop the lifespan and various characteristics when compared to the normal installation.
- When using in a space below the insulation distance below, damage may be caused by the switching arc.

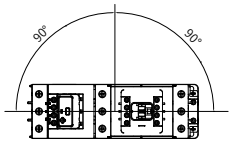
Installation

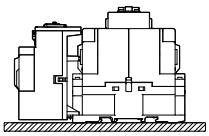
General Attachment



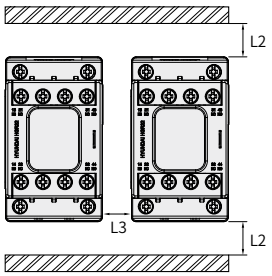


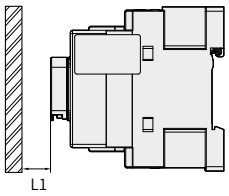
Transverse and Horizontal Attachments





Insulation Distance





※ Continuous energization may reduce coil life due to increase in temperature. If the insulation distance is not maintained, external discoloration may occur due to coil heat.

Unit : mm

Product / Distance	HGR	HGC9 ~ 100	HGC115 ~ 150	HGC185 ~ 265	HGC300 ~ 500	HGC630 ~ 800
L1	5	5	15	15	15	20
L2	20	20	30	50	50	80
L3	2	2	10	10	10	10

Handling Precautions

Caution

- Do not store and use in abnormal environment such as high temperature, high humidity, dust, corrosive gas, vibration, impact and others It may cause electric shock, fire and malfunction.
- Ensure that rubbish, foreign substances such as concrete powder, metal powder, rainwater and others do not flow in.
- When handling the product, do not use lubricants. It may cause electric shock and fire.
- Electrical works must be conducted by qualified technicians with certifications related to electrical works. It may cause electric shock and fire.

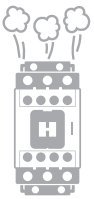
Inspection Items Before Use

- B˙Ensure that calibration work is not skipped. It may cause malfunction.

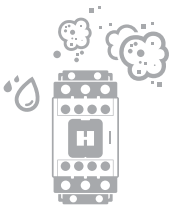
Installation, Usage and Maintenance Precautions

- C˙Frequently check the fastening state, assembly and combination status of the bolt. It may cause electric shock, fire and malfunction.
- Check whether the rated current, voltage and frequency conform to the intended specification. It may cause electric shock, fire and malfunction.
- As for the wiring work, check whether the upper circuit breaker has been turned off so that electricity does not flow. It may cause electric shock.
- Connect the power that conforms to the ratings of the switch/contactor's main unit. It may cause fire and malfunction.
- As for the wiring and terminal, use the standard product and as for the terminal screw, fasten using the torque indicated in the user manual. It may cause fire.
- Frequently check the wiring state and shape. It may cause electric shock, fire and malfunction.
- Check whether parts have been omitted and replace damaged products and parts. It may cause electric shock, fire and malfunction.
- In case of abnormality in the contact, replace immediately and do not arbitrarily process such as trimming the contact using string or use lubricant. It may cause electric shock, fire and malfunction
- Use the specified tool. It may cause damage and malfunction.

Installation Precautions



- **Do not block the arc gas exhaust hole**
  It may drop the breaking capacity.

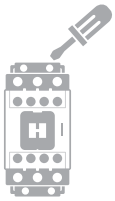


- **Pay attention to dust, metal scraps and others**
  After installation, cover the protection cover while working with the machine



- **Do not remove the insulation plate attached at the bottom of the circuit breaker.**
  It may destroy the insulation and the insulation performance.

Connection Precautions



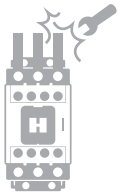
- **When fastening the terminal screw, it should be fastened according to the specified torque.**
  Incomplete fastening of the terminal screw may cause overheating so each terminal screw must be fastened completely according to the specified torque.
  In addition, excessive fastening torque may cause damage in the terminal screw and the circuit breaker case.



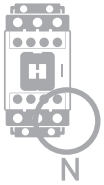
- **Use of lubricant at the terminal screw part is prohibited**
  Lubricant reduces the friction of the screw, causing the screw to loosen, ultimately leading to an increase in temperature.



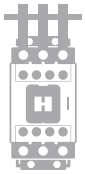
- **Exposed conductor must be insulated**
  Insulating tube or insulating tape must be used to complete insulation between the bare conductors of the MC. In case the terminals are not insulated, it may cause secondary shot-circuit during short-circuit accidents.



- **STUD must not be deformed**
  Excessive force must not be applied to the stud at the conductor connecting part of the rear connection type. In addition, stud must not be deformed during wiring.



- **In case of 4 pole circuit breaker, the neutral wire of 3 phase 4 wire must be connected to the N phase.**
  It may not function in overcurrent which may cause fire.



- **The conductor must be fixed firmly on a flat state**
  As for the connecting conductor, electromagnetic force between conductors is generated by extremely big fault current so it must be fixed firmly.

Acquired Standards

Approvals & Certificates

MS

● : Acquired
◎ : In Progress (Expected)

Type of Certification	Approval		Certificate
Type of Standard	Safety Certification	IEC	IEC
Mark			
Testing Institute	KETI	-	KTC
Certification Country	Korea	Europe	Korea
HGC9/12/18/22B 10/01	●	●	●
HGC9/12/18/22B 11	●	●	◎
HGC25/32/40B 11	●	●	◎
HGC50/65B 22	●	●	●
HGC75/85/100B 22	●	●	●

Essential for Today
Potential for Tomorrow

HD HYUNDAI ELECTRIC

Korea			
Head Office	477, Bundang Suseo-ro, Bundang-gu, Seongnam-si, Korea	Tel: +82-2-479-9180	Fax: +82-2-500-4958
Ulsan Factory	700 Bangeojinsunhwan-doro, Dong-gu, Ulsan, Korea	Tel: +82-52-202-8114	Fax: +82-52-202-8010
Seonam Factory	223, Sapyeong-ro, Nam-gu, Ulsan, Korea		
Cheongju Distribution Campus	20, Guksasandan 1-ro, Oksan-myeon, Heungdeok-gu, Cheongju-si, Chungcheongbuk-do, Korea		
Reliability Center	17-10, Mabuk-ro 240beon-gil, Giheung-gu, Yongin-si, Gyeonggi-do, Korea	Tel: +82-31-289-5000	Fax: +82-31-289-5040
Overseas Subsidiaries			
Atlanta(U.S.A)	2405 Satellite Blvd, Duluth GA 30096, U.S.A.	Tel: +1-770-377-7370	
Alabama(U.S.A)	215 Folmar Parkway, Montgomery, AL 36105, U.S.A	Tel: +1-334-481-2000	Fax: +1-334-604-2498
Texas(U.S.A)	14114 Pkwy Ste 350 Dallas, TX 75254	Tel: +1-214-502-0221	
Frankfurt(Germany)	HD Hyundai Electric Europe GmbH Düsseldorfer Str. 40 (Eingang 2, 8G.), 65760, Eschborn, Germany	Tel: +49-697-116-3593	
Riyadh	Office No. 529, 5th floor Akaria-3 Building, Olaya Street, PO Box 9187, Riyadh 11413, Kingdom of Saudi Arabia	Tel: +966-11-419-0168	Fax: +966-11-4190-0220
Yangzhong(China)	No.9, Xiandai Road, Xinba Scientific and Technologic Zone, Yangzhong, Jiangsu, P.R.C. Zip:212212, China	Tel: +86-511-8842-0510	Fax: +86-511-8842-0388
Branch Offices			
London(U.K)	50-68 High Street, Weybridge, KT13 8BL, United Kingdom	Tel: +44-(0)20-8600-7122	
Russia(Moscow)	World Trade Center, Ent.6, #412, Krasnopresnenskaya Nab.12, Moscow, 123610, Russia	Tel: +7-495-258-1381	
Riyadh	Office No. 530A, 5th Floor Akaria-2 Bldg., Olaya St., Short Address RHOB6847, Riyadh 12244, Kingdom of Saudi Arabia	Tel: +966-11-210-1107	Fax: +966-11-210-1107
U.A.E(Dubai)	Unit 205, Emaar Square Building NO.4 Sheikh Zayed Road, Pobox 252458, Dubai, U.A.E	Tel: +971-50-771-0378	Fax: +971-4-425-7996
Vietnam(Ho Chi Minh)	Room 1405, 14th Floor Centec Tower, 72-74 Nguyen Thi Minh Khai Street Vo Thi Sau Ward, District 3, Ho Chi Minh City, Vietnam	Tel: +84-28-3535-0465	
Japan(Osaka)	1-Room 5th Floor Nagahori Plaza Bldg. 2-4-8 Minami Senba, chuo-ku, Osaka 542-0081, Japan	Tel: +81-06-6261-5766~7	Fax: +81-06-6261-5818
R&D Centers			
Switzerland(Zurich)	Hardturmstrasse 135, CH-8005, Zurich, Switzerland	Tel: +41-44-527-0056	
Hungary(Budapest)	H-1145 Bacskai utca 29/a, Budapest, Hungary	Tel: +36-1-273-3733	
China(Shanghai)	Room 10102, Building 10, 498 Guoshoujing Road, Pudong, Shanghai, China(201203)	Tel: +86-21-5013-3393 #139	
Customer Service	service@hd.com		